Politecnico di Milano

**Machine Learning for Mechanical Systems**

# PPO-Based Control of an Active Motion Surface for Parcel Singulation


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**Abstract**

This project investigates Proximal Policy Optimization for the continuous control of a $5 \times 5$ Active Motion Surface devoted to longitudinal parcel singulation. The resulting Reinforcement Learning problem comprises a 100-dimensional grid-based observation and 50 normalized continuous actions controlling the velocity magnitude and direction of the AMS cells. The complete workflow was implemented in MATLAB using Reinforcement Learning Toolbox.

Before training, the numerical environment was systematically verified with respect to observation and action ordering, action denormalization, parcel generation, kinematics, boundary handling, collision correction, reproducible reset, and episode termination. Particular attention was devoted to separating the shaped reward used for policy learning from the physical metrics used for performance assessment. Final singulation was evaluated from the longitudinal edges of consecutive parcels, using a strict clearance criterion of $g > 5\,\mathrm{mm}$, rather than centroid distance alone.

A reference PPO configuration was compared with configurations identified through a high-fidelity Bayesian Optimization campaign. The shortlisted candidates were retrained and evaluated under a common budget of 2000 training episodes and 100 nominal evaluation episodes. The selected configuration, `BO_014`, achieved a mean strict edge-gap accuracy of 99.78 %, an episode-level standard deviation of 1.56 percentage points, and a mean maximum overlap depth of 1.33 mm.

The selected hyperparameter configuration was subsequently retrained with four independent random seeds. The resulting agents achieved a multiseed mean accuracy of 97.861 %, with a between-training-seed standard deviation of 2.217 percentage points, confirming that the result was not limited to one favourable training realization.

An evaluation-only stress campaign showed good generalization to a doubled parcel count and to rectangular parcels within the tested dimensional range. The main degradation occurred under denser inlet conditions: reducing the generation period to 0.50 s lowered mean accuracy to 85.22 %, while a feasible three-parcel inlet configuration produced 78.34 %. All evaluated episodes reached completion, although completion alone did not guarantee successful singulation.

Overall, the study demonstrates that PPO can provide accurate and repeatable parcel singulation within the verified kinematic simulation. The principal remaining limitation is generalization to dense and non-pairwise inlet structures, while transfer to a physical AMS would require higher-fidelity parcel dynamics, perception uncertainty, actuator dynamics, and experimental validation.

# Contents

# 1 Introduction

## 1.1 Motivation and project scope

Parcel singulation is the process of converting a batch or compact flow of parcels into an ordered stream with sufficient longitudinal clearance between consecutive items. This operation is relevant in intralogistics because downstream tasks such as identification, routing, and automated handling generally require the parcels to be processed individually rather than as mutually overlapping or closely packed objects [1].

The system considered in this project is based on an Active Motion Surface (AMS), namely a matrix of locally controllable cells that can modify the motion of the parcels travelling across it. The implemented AMS contains $5 \times 5$ cells, each providing a velocity-magnitude command and a velocity-direction command. Consequently, the controller must determine 50 continuous action components while coordinating several parcels with variable dimensions and inlet positions.

Rather than prescribing a fixed sequence of actuator commands for every possible parcel configuration, this work investigates a control policy learned through Reinforcement Learning (RL). The objective is to determine the AMS commands that promote positive longitudinal separation before the parcels reach the final downstream evaluation line. Proximal Policy Optimization (PPO) is employed as the learning algorithm. PPO is an on-policy actor–critic method whose clipped surrogate objective limits excessively large policy updates during training [2]. The complete simulation, training, and evaluation workflow is implemented in MATLAB using Reinforcement Learning Toolbox.

## 1.2 Methodological positioning and contributions

The work by Roveda et al. constitutes the main methodological reference for the application of PPO to AMS-based parcel sorting [1]. The reference framework combines a kinematic AMS model, collision handling, offline PPO training, machine-vision-based parcel tracking, and experimental deployment on a physical sorting module. The present project adopts the general AMS control problem and the use of PPO, but it is not a reproduction of the complete framework presented in that study.

The scope of this report is limited to the supplied MATLAB simulation environment and to the control and evaluation choices developed for the course project. In particular, the implemented system uses a project-specific observation, reward, episode-termination rule, downstream region, and physical evaluation protocol. Machine vision and hardware deployment are outside the scope of the present work. Therefore, all conclusions reported here refer to the verified behaviour of the implemented numerical environment and of the trained policies, rather than to experimental performance on the physical AMS.

The main contributions of the project are:

- the systematic verification of the implemented environment, including parcel generation, kinematics, action ordering and scaling, observation construction, boundary handling, collision correction, episode termination, and visual consistency;
- the definition of a downstream edge-gap reward for policy learning, together with independent edge-based metrics for the physical evaluation of parcel singulation;

- the establishment of a reproducible PPO baseline and of a common nominal protocol for training and post-training evaluation;
- the optimization and equal-budget comparison of selected PPO hyperparameters through Bayesian Optimization, leading to the selection of a final nominal policy configuration;
- the assessment of training stochasticity through independent PPO retraining runs with multiple random seeds;
- an evaluation-only stress campaign in which the saved policies are tested under non-nominal parcel counts, inlet rates, downstream velocities, parcel geometries, and batch structures without modifying the policy or retraining the agent.

A central methodological principle throughout the project is the separation between the reward used to train the agent and the physical Key Performance Indicators (KPIs) used to judge the resulting policy. Training reward is therefore treated as a learning diagnostic, while final conclusions are based on longitudinal parcel-edge gaps and on complementary operational metrics.

### 1.3 Report organization

The remainder of the report is organized as follows. Section 2 describes the physical system, the parcel-flow model, and the RL observation and action spaces. Section 3 introduces the training reward and the independent physical performance metrics. Section 4 presents the PPO architecture, training configuration, and experimental methodology. Section 5 discusses the reference baseline, the hyperparameter-optimization campaign, and the selection of the final nominal policy. Section 6 evaluates sensitivity to the training seed and performance under non-nominal operating conditions. Section 7 discusses the observed behaviour, methodological limitations, and possible improvements. Finally, Section 8 summarizes the main findings and identifies the most direct continuation of the present work.

## 2 Physical System and Reinforcement-Learning Formulation

This section reconstructs the nominal environment implemented in `RL_environment_v12.m`. The description is based on the effective MATLAB implementation rather than on an assumed equivalence with the reference framework of Roveda et al. [1]. The resulting task is formulated as an episodic continuous-action Reinforcement Learning problem in which the agent observes the parcels located on the Active Motion Surface and commands the local motion of its 25 cells.

### 2.1 Active Motion Surface and parcel flow

The simulated system consists of a square Active Motion Surface (AMS) followed by a passive downstream treadmill. The AMS is composed of $5 \times 5$ square cells with side length

$$d_{\text{AMS}} = 0.2\,\text{m},$$

and therefore covers a surface of $1\,\text{m} \times 1\,\text{m}$. Each cell provides two independent commands: the magnitude and the direction of the velocity imposed on the parcel whose centroid occupies that cell. The complete AMS consequently requires 50 continuous action components.

The environment is divided into two longitudinal regions:

$$0 \le y \le 1\,\mathrm{m} \qquad \text{controllable AMS region,} \tag{1}$$

$$1\,\mathrm{m} < y \le 2\,\mathrm{m} \qquad \text{passive downstream region.} \tag{2}$$

The AMS exit and the final evaluation line are consequently defined as

$$y_{\mathrm{exit}} = 1\,\mathrm{m}, \qquad y_{\mathrm{eval}} = 2\,\mathrm{m}. \tag{3}$$

Inside the first region, parcel motion depends on the commands selected by the PPO policy. Once the parcel centroid crosses the AMS exit, the local AMS commands no longer affect that parcel and the downstream treadmill transports it at the constant nominal velocity

$$v_{\mathrm{treadmill}} = 1.9\,\mathrm{m/s}.$$

The one-metre downstream region is therefore not an additional controlled AMS module: it is a passive transport section in which the consequences of the previous AMS commands become observable.

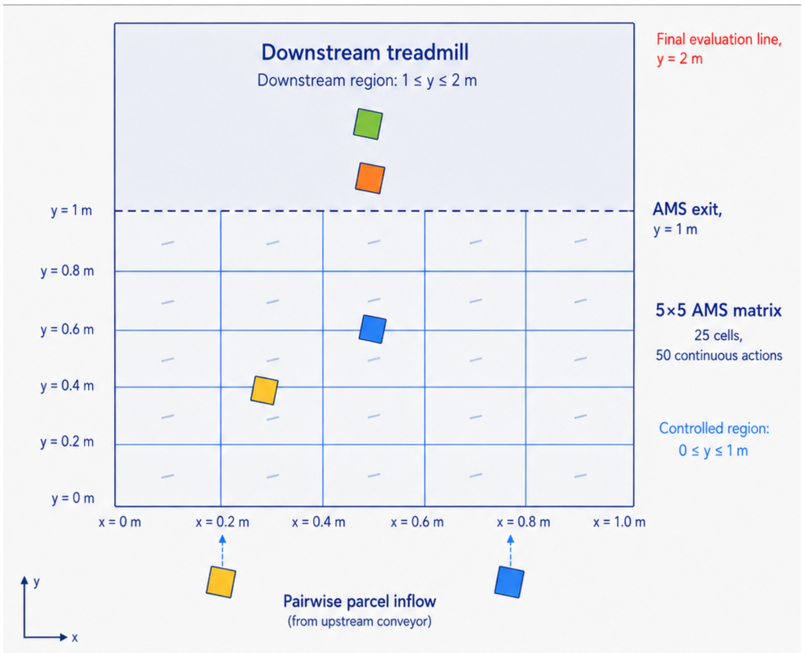


**Figure 1:** Nominal parcel-singulation environment. The PPO policy controls the $5 \times 5$ AMS in $0 \le y \le 1\,\mathrm{m}$, whereas parcels are transported by the passive downstream treadmill between the AMS exit and the final evaluation line at $y = 2\,\mathrm{m}$.

The principal physical and temporal parameters of the nominal environment are summarized in Table 1.

**Table 1:** Nominal physical and temporal parameters of the implemented environment.

| Parameter | Value | Description |
|---|---|---|
| AMS matrix | $5 \times 5$ | 25 locally controlled cells |
| AMS cell side | $0.2\,\text{m}$ | Total AMS dimensions of $1\,\text{m} \times 1\,\text{m}$ |
| Simulation timestep | $0.01\,\text{s}$ | Discrete kinematic update interval |
| Downstream velocity | $1.9\,\text{m/s}$ | Constant passive treadmill velocity |
| Total parcel count | 10 | Five two-parcel inlet batches |
| Batch size | 2 parcels | One parcel initialized in each lateral half of the inlet |
| Generation period | $0.75\,\text{s}$ | Interval between successive batches after reset |
| Nominal parcel geometry | Square | Axis-aligned footprint in the $x$-$y$ plane |
| Parcel side length | $0.05\,\text{m}$ to $0.40\,\text{m}$ | Independently randomized at episode reset |

## 2.2 Kinematic environment model

Each nominal parcel $k$ is represented by its centroid coordinates

$$\boldsymbol{p}_k(t) = \begin{bmatrix} x_k(t) & y_k(t) \end{bmatrix}^T$$

and by its side length $d_k$. Mass, inertia, friction, parcel rotation, and actuator dynamics are not included in the environment state. The implemented model is therefore purely kinematic, consistently with the general modeling approach adopted in the methodological reference [1].

AMS-cell membership is determined from the parcel centroid. Let $v_{ij}(t)$ and $\theta_{ij}(t)$ denote the physical velocity magnitude and direction assigned to cell $(i, j)$. If the centroid of parcel $k$ lies in that cell, its position is updated as

$$x_k(t + \Delta t) = x_k(t) + v_{ij}(t)\Delta t \sin\theta_{ij}(t), \tag{4}$$

$$y_k(t + \Delta t) = y_k(t) + v_{ij}(t)\Delta t \cos\theta_{ij}(t), \tag{5}$$

where

$$\Delta t = 0.01\,\text{s}.$$

With this convention, $\theta_{ij} = 0$ corresponds to motion along the positive $y$-direction, whereas positive and negative angles introduce lateral motion along positive and negative $x$, respectively.

The transition from active to passive transport is also centroid-based. When

$$y_k > y_{\text{exit}},$$

the parcel is no longer assigned to an AMS cell and its motion becomes

$$x_k(t + \Delta t) = x_k(t), \tag{6}$$

$$y_k(t + \Delta t) = y_k(t) + v_{\text{treadmill}}\Delta t. \tag{7}$$

Consequently, the downstream region preserves the lateral coordinate reached at the AMS exit and applies the same longitudinal velocity to every parcel.

The lateral boundaries are enforced using the complete parcel footprint rather than only its centroid. After actuation and collision correction, the coordinate is constrained according to

$$\frac{d_k}{2} \le x_k \le 1\,\mathrm{m} - \frac{d_k}{2}, \tag{8}$$

which prevents any portion of a nominal square parcel from leaving the physical width of the conveyor.

Parcel interactions are represented through a geometric collision-correction logic inherited from the supplied MATLAB baseline. When the implemented contact conditions detect a transition into footprint overlap along either $x$ or $y$, the penetration is divided between the two parcels and their centroids are displaced in opposite directions. The correction also includes the implemented contact tolerance

$$\varepsilon_{\text{contact}} = 5\,\mathrm{mm}.$$

This procedure is not a force-based impact model and does not predict contact forces or impulses. A recorded collision correction must therefore be interpreted as a numerical overlap-resolution event rather than as a complete physical description of a parcel collision.

## 2.3 Observation space

The policy does not receive the complete simulator state or a variable-length list of all generated parcels. Instead, it receives a fixed-size grid-centric observation constructed from the $5 \times 5$ AMS matrix.

Cells are scanned in row-major order. Denoting the row index along $y$ by $i$ and the column index along $x$ by $j$, the scalar cell index is

$$c = j + (i-1)n_j, \qquad n_j = 5. \tag{9}$$

Thus, cells 1–5 form the first row at the inlet, ordered from left to right; cells 6–10 form the second row; and cells 21–25 form the final row adjacent to the AMS exit.

For an occupied cell $c$, the corresponding four-dimensional feature block is

$$\boldsymbol{o}_c = \begin{bmatrix} x_{\text{rel},c} \\ y_{\text{rel},c} \\ d_{\text{norm},c} \\ d_{\text{norm},c} \end{bmatrix}, \tag{10}$$

where

$$x_{\text{rel},c} = \frac{x_k - x_c^{\text{origin}}}{d_{\text{AMS}}}, \tag{11}$$

$$y_{\text{rel},c} = \frac{y_k - y_c^{\text{origin}}}{d_{\text{AMS}}}, \tag{12}$$

$$d_{\text{norm},c} = \frac{d_k}{2d_{\text{AMS}}}. \tag{13}$$

The local position components belong to $[0, 1]$, while the nominal parcel size range produces

$$d_{\text{norm},c} \in [0.125, 1].$$

Because nominal parcels are square, the same normalized side length is stored in both size channels. Empty cells are represented by four zeros.

The complete observation is obtained by stacking the 25 cell blocks:

$$\boldsymbol{o}_t = \begin{bmatrix} \boldsymbol{o}_1^T & \boldsymbol{o}_2^T & \cdots & \boldsymbol{o}_{25}^T \end{bmatrix}^T \in \mathbb{R}^{100}. \tag{14}$$

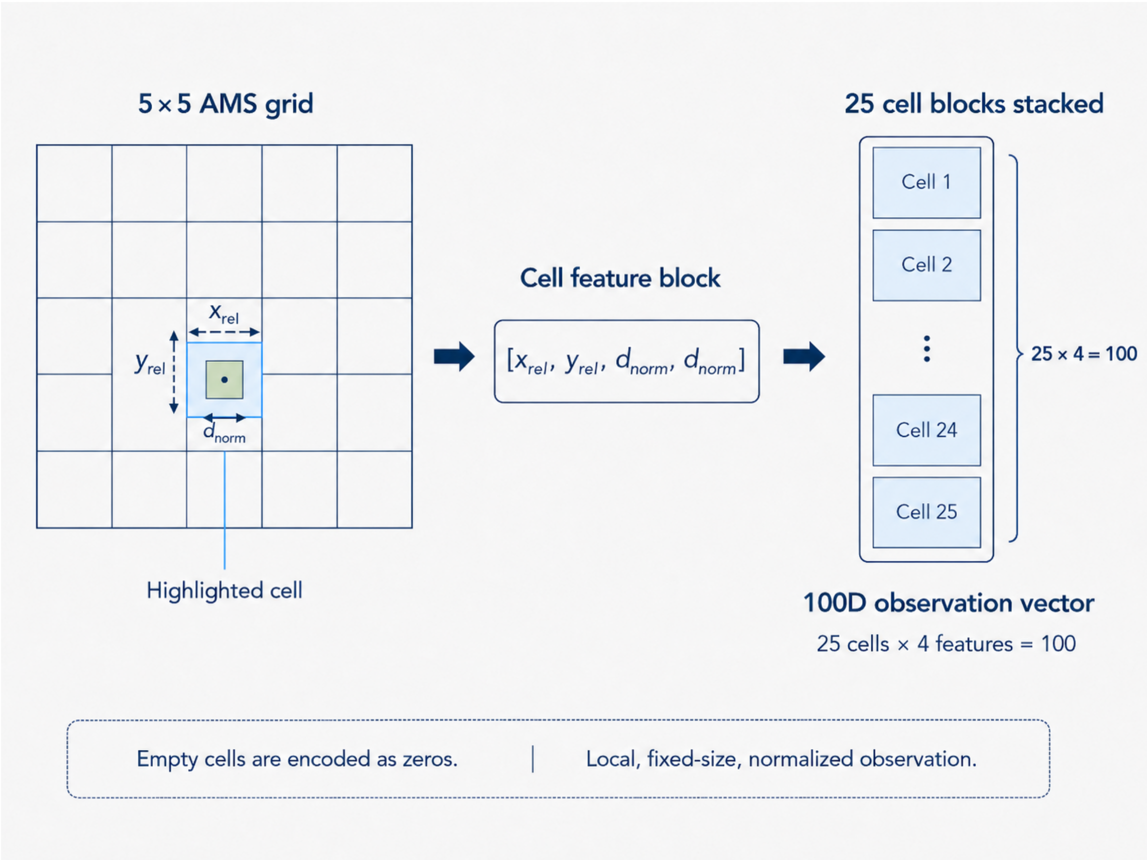


**Figure 2:** Construction of the 100-dimensional grid-centric observation. Each AMS cell contributes the normalized local parcel position and two identical normalized size components; empty cells contribute zero blocks.

Cell membership is based on the parcel centroid. If more than one parcel centroid lies inside the same cell, the implemented scan stores only the first active parcel found in parcel-index order. Moreover, parcels that have already entered the downstream region are omitted from the observation.

The policy input is therefore an observation rather than the full Markov state of the simulator. In particular, the agent does not directly observe the final downstream gaps, parcel velocities, previous observations, or the complete set of parcels when multiple objects occupy the same cell. The effect of an AMS command on downstream separation is consequently communicated to the policy through delayed rewards during training.

### 2.4 Action space

The actor produces a normalized continuous action vector with 50 components:

$$\boldsymbol{a}_t = \begin{bmatrix} a_{\theta,1} & a_{v,1} & a_{\theta,2} & a_{v,2} & \cdots & a_{\theta,25} & a_{v,25} \end{bmatrix}^T, \qquad \boldsymbol{a}_t \in [-1,1]^{50}. \tag{15}$$

The cell ordering is the same row-major ordering used for the observation. For each cell $c$, the first component controls direction and the second controls velocity magnitude.

Before application to the environment, every normalized action is clipped to $[-1, 1]$ and converted to its physical interval through the affine mapping

$$u = u_{\min} + \frac{a+1}{2}\left(u_{\max} - u_{\min}\right). \tag{16}$$

The resulting physical commands satisfy

$$\theta_c \in \left[-\frac{\pi}{4}, \frac{\pi}{4}\right], \tag{17}$$

$$v_c \in [0.5\,\mathrm{m/s}, 2.2\,\mathrm{m/s}]. \tag{18}$$

More explicitly,

$$\theta_c = \frac{\pi}{4} a_{\theta,c}, \tag{19}$$

$$v_c = 1.35\,\mathrm{m/s} + 0.85\,\mathrm{m/s}\, a_{v,c}. \tag{20}$$

A zero normalized command therefore corresponds to straight longitudinal motion at $1.35\,\mathrm{m/s}$, rather than to zero physical velocity.

At each timestep the policy specifies commands for all 25 cells, including currently empty cells. Only the pair of commands associated with the cell occupied by a parcel centroid affects that parcel during the current transition.

### 2.5 Episode initialization and termination

At environment reset, the side lengths of all ten nominal parcels are sampled independently from a uniform distribution over

$$d_k \in [0.05\,\mathrm{m}, 0.40\,\mathrm{m}].$$

The first pair is immediately placed close to the inlet line $y = 0$. One parcel is initialized in the left half of the AMS width and the other in the right half. Small randomized longitudinal offsets and geometric corrections are applied to avoid invalid initial positions and excessive footprint overlap.

Four additional two-parcel batches are subsequently introduced at intervals of $0.75\,\mathrm{s}$, until the nominal total of ten parcels is reached. Each new batch follows the same left–right initialization logic. Parcel generation is therefore stochastic in both size and inlet position, whereas the number of parcels, batch cardinality, generation period, and geometric limits remain fixed in the nominal environment.

The environment uses the MATLAB global random-number generator. Reproducible initializations are obtained by setting the random seed in the external training or evaluation script before the environment is reset. The environment itself does not introduce a separate uncontrolled random stream.

The episode does not terminate when a parcel centroid reaches the final evaluation line. Termination is enabled only after all ten parcels have been generated and the leading edge of every parcel has reached $y_{\text{eval}}$. For a square parcel $k$, the condition is

$$y_k + \frac{d_k}{2} \geq y_{\text{eval}}. \tag{21}$$

The global terminal condition is therefore

$$\text{IsDone} = (N_{\text{generated}} = 10) \wedge \bigwedge_{k=1}^{10} \left( y_k + \frac{d_k}{2} \geq y_{\text{eval}} \right). \tag{22}$$

Using the parcel edge prevents the episode logic from depending only on centroid positions and makes the termination condition consistent with the geometric representation of the parcels.

Before PPO training was interpreted, the observation dimensions and ordering, action denormalization, parcel kinematics, inlet generation, lateral boundaries, collision corrections, reset reproducibility, and front-edge termination were checked numerically and through visual rollouts. All subsequent nominal results refer to this verified environment configuration.

# 3 Training Reward and Physical Performance Metrics

The learning signal used to optimize the PPO policy must be distinguished from the quantities used to assess whether parcel singulation has been physically achieved. The training reward is an instantaneous, shaped signal evaluated repeatedly while parcels travel through the downstream region. The final performance metrics instead describe the longitudinal configuration of the complete parcel stream at the end of an evaluation episode.

This distinction is essential throughout the present work. Training reward is used to guide policy updates and to diagnose the evolution of the learning process, whereas policy selection and physical conclusions are based on independently computed parcel-edge metrics.

## 3.1 Downstream training reward

The reward is evaluated in the passive downstream region introduced in Section 2.1. At timestep $t$, let $\mathcal{P}_t$ denote the set of parcels whose centroids satisfy

$$y_{\text{exit}} \leq y_k(t) \leq y_{\text{eval}}. \tag{23}$$

The membership criterion is centroid-based: a parcel contributes to the reward whenever its centroid lies between the AMS exit and the final evaluation line.

Let

$$K_t = |\mathcal{P}_t|$$

be the number of parcels currently present in this region. If $K_t < 2$, no pairwise longitudinal gap can be defined and the immediate reward is set to

$$r_t = 0. \tag{24}$$

When at least two parcels are available, they are sorted according to their centroid coordinate:

$$y_{(1)} \leq y_{(2)} \leq \cdots \leq y_{(K_t)}. \tag{25}$$

The first parcel in this ordering is the rearmost one, whereas the last parcel is the most advanced along the positive $y$-direction.

For two consecutive parcels $k$ and $k+1$, the signed longitudinal edge gap is

$$g_{t,k} = \left( y_{(k+1)} - \frac{d_{(k+1)}}{2} \right) - \left( y_{(k)} + \frac{d_{(k)}}{2} \right), \qquad k = 1, \dots, K_t - 1. \tag{26}$$

A positive value indicates clearance between the two parcel footprints, while a negative value indicates longitudinal overlap.

Each gap is converted into the local score

$$s_{t,k} = \min\left[10K_t \arctan\left(g_{t,k}\right) + K_t,\ 6\right]. \tag{27}$$

This functional form follows the reward design proposed by Roveda et al. [1]. The dependence on $K_t$ changes both the offset and the sensitivity of the score according to the number of parcels currently present in the rewarding region. The upper saturation at 6 limits the benefit obtained by indefinitely increasing an already positive gap. No corresponding lower saturation is applied, allowing sufficiently large overlaps to generate negative local scores.

The base instantaneous reward is the minimum of the local scores:

$$r_t^{\text{base}} = \min_{1 \leq k \leq K_t - 1} s_{t,k}. \tag{28}$$

Consequently, the most critical current gap governs the learning signal. This worst-gap aggregation discourages a policy from producing large separation for some parcels while leaving another adjacent pair in severe overlap.

If at least three parcels are present and all current gaps are strictly positive,

$$K_t \geq 3 \qquad \text{and} \qquad g_{t,k} > 0 \quad \forall k, \tag{29}$$

an additional bonus is applied:

$$r_t = r_t^{\text{base}} + 20. \tag{30}$$

Otherwise,

$$r_t = r_t^{\text{base}}.$$

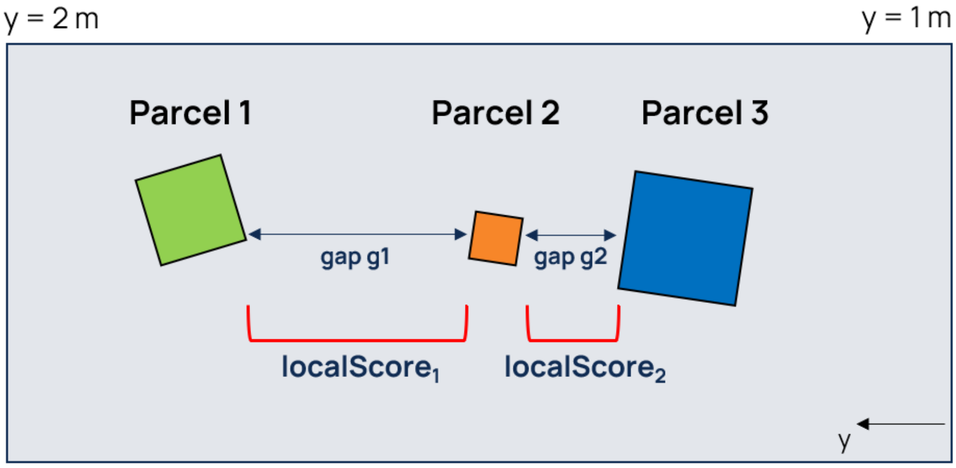


**Figure 3:** Construction of the downstream training reward. Parcels are ordered along the longitudinal direction, an edge gap and a local score are computed for each consecutive pair, and the minimum local score defines the base instantaneous reward.

The reward is deliberately shaped and temporally dense, but it is not a direct measure of final task success. In particular:

- the same physical gap produces different scores for different values of $K_t$;
- at $g_{t,k} = 0$, the local score is $\min(K_t, 6)$, and is therefore not generally zero;
- the $+20$ bonus requires only positive gaps and does not enforce the stricter 5 mm evaluation threshold;
- the reward is accumulated repeatedly while parcels remain in the downstream region;
- the minimum operator describes the worst instantaneous pair, not the complete final output stream;
- the numerical return depends on the sequence of downstream occupancies and on the episode duration.

Moreover, the downstream parcels used to compute the reward are not included in the observation described in Section 2.3. The policy must therefore associate its earlier AMS commands with their delayed effects on downstream separation.

For these reasons, *AverageReward* and the episode return are treated only as training diagnostics. They are not used as sufficient evidence of successful parcel singulation.

### 3.2 Edge-based singulation criterion

Physical singulation is evaluated from the boundaries of consecutive parcels, rather than from their centroid distance. Consider two parcels ordered along the longitudinal direction, with a rear parcel $r$ and a front parcel $f$. Their relevant longitudinal edges are

$$y_{\mathrm{top},r} = y_r + \frac{d_r}{2}, \tag{31}$$

$$y_{\mathrm{bottom},f} = y_f - \frac{d_f}{2}. \tag{32}$$

The signed physical gap is

$$g = y_{\text{bottom},f} - y_{\text{top},r}. \tag{33}$$

Its interpretation is

$$\begin{cases} g > 0, & \text{positive physical clearance,} \\ g = 0, & \text{parcel boundaries in longitudinal contact,} \\ g < 0, & \text{longitudinal footprint overlap.} \end{cases} \tag{34}$$

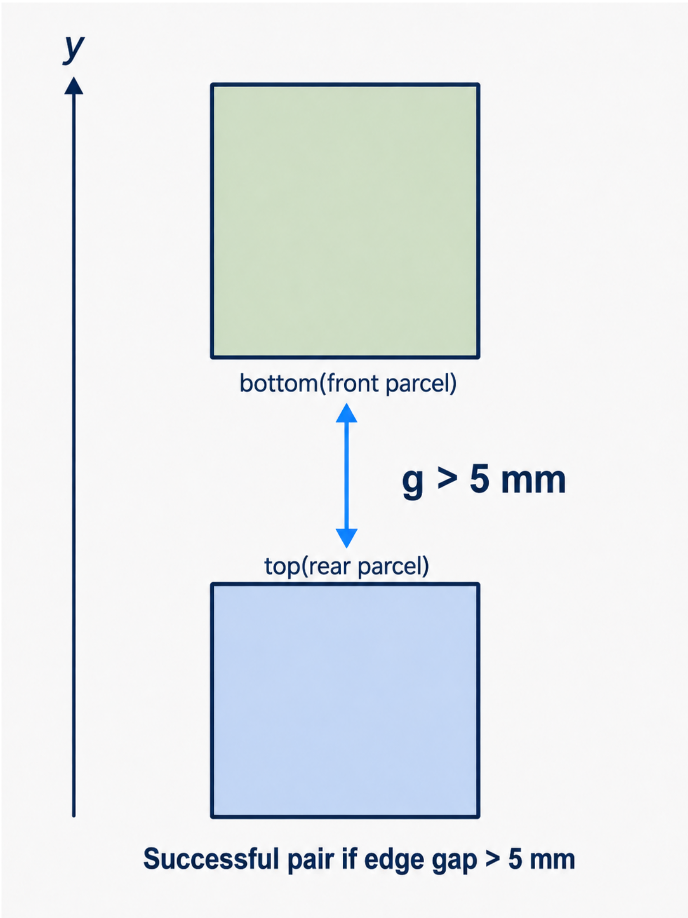


**Figure 4:** Definition of the longitudinal edge gap between two consecutive parcels. The distance is measured between the bottom edge of the front parcel and the top edge of the rear parcel; centroid separation alone is not used to declare successful singulation.

The edge-based definition is necessary because the parcels have variable dimensions. For example, a positive centroid distance does not guarantee physical separation if the sum of the two parcel half-lengths exceeds that distance. In general,

$$g = (y_f - y_r) - \frac{d_f + d_r}{2}, \tag{35}$$

which explicitly shows why centroid distance alone is insufficient.

Two separation thresholds are retained:

$$g > 0 \qquad \text{permissive physical-separation criterion,} \tag{36}$$

$$g > \delta_g, \qquad \delta_g = 5\,\text{mm} \qquad \text{strict project evaluation criterion.} \tag{37}$$

The first criterion only requires the parcel footprints not to overlap or touch. The second introduces a positive clearance margin and constitutes the primary criterion used for policy comparison in the final study. The 5 mm value is a project-specific evaluation threshold and should not be interpreted as a universal industrial singulation requirement.

For a stream of $N$ parcels, the parcels are sorted by $y$ and $N_{\text{pair}} = N - 1$ consecutive gaps are evaluated. The accuracy associated with threshold $\delta$ is

$$A_\delta = \frac{100}{N-1} \sum_{k=1}^{N-1} \mathbb{I}\left(g_k > \delta\right), \tag{38}$$

where $\mathbb{I}(\cdot)$ is the indicator function.

The word *pair* in this definition refers to two adjacent parcels in the final longitudinally ordered stream. It does not refer to the two parcels that were introduced together during a generation event.

In the nominal ten-parcel environment, each completed episode contains nine evaluated gaps. Individual episode accuracies are therefore quantized in increments of

$$\frac{100}{9} \approx 11.11\,\%.$$

Mean accuracies calculated over multiple episodes are not subject to this quantization.

The final metric is computed from the complete parcel configuration at the end of the rollout. For a normally completed episode, this occurs after the front edge of every generated parcel has reached the final evaluation line, according to the termination condition defined in Section 2.5. The implementation does not evaluate only the pair nearest to the line, nor does it associate success exclusively with the original inlet batches.

### 3.3 Primary and secondary evaluation metrics

The strict final edge-gap accuracy,

$$A_{5\,\text{mm}} = \frac{100}{N-1} \sum_{k=1}^{N-1} \mathbb{I}\left(g_k > 5\,\text{mm}\right), \tag{39}$$

is the primary physical metric used to compare PPO configurations. It directly measures the fraction of consecutive parcel pairs that satisfy the selected clearance requirement in the final output stream.

A single percentage is nevertheless insufficient to characterize policy behaviour completely. The complete evaluation set is summarized in Table 2.

**Table 2:** Physical and operational metrics used for policy evaluation.

| Category | Metric | Interpretation |
|---|---|---|
| Primary physical metric | Final edge-gap accuracy $g > 5\,\text{mm}$ | Percentage of consecutive pairs satisfying the strict clearance criterion in the final parcel configuration. |
| Secondary final metric | Final edge-gap accuracy $g > 0$ | Percentage of pairs with strictly positive physical clearance, without the additional 5 mm margin. |
| Secondary final metric | Final mean edge gap | Arithmetic mean of all final consecutive edge gaps. It describes average stream clearance but can conceal one critical pair. |
| Secondary final metric | Final minimum edge gap | Smallest final gap in the episode. It identifies the most critical pair and preserves the sign of the gap. |
| Secondary final metric | Final maximum overlap depth | Largest longitudinal penetration in the final stream, defined as $\max(0, -\min_k g_k)$. |
| Temporal diagnostic | Time-averaged edge-gap accuracy | Fraction of valid pair observations satisfying the selected threshold while parcels occupy the downstream region. |
| Temporal diagnostic | Time-averaged mean edge gap | Mean of all edge gaps accumulated over valid downstream evaluation steps. |
| Operational diagnostic | Completion time and number of steps | Time required to satisfy the front-edge termination condition. |
| Operational diagnostic | Equivalent throughput | Number of generated parcels divided by the episode completion time and converted to parcels per hour. |
| Operational diagnostic | Collision-correction count | Number of numerical overlap-resolution events accumulated during the rollout. It is not a force or impact measurement. |
| Robustness diagnostic | Non-completion rate | Fraction of evaluation episodes that fail to reach the termination condition within the prescribed maximum number of steps. |

The final maximum overlap depth is defined as

$$o_{\max} = \max\left(0, -\min_k g_k\right). \tag{40}$$

Thus, $o_{\max} = 0$ when no final gap is negative, while a positive value reports the depth of the worst residual longitudinal overlap.

The equivalent episode throughput is calculated as

$$Q_{\text{episode}} = \frac{N_{\text{generated}}}{T_{\text{completion}}} 3600, \tag{41}$$

and is expressed in parcels per hour. Because the nominal rollout contains a finite transient sequence of ten parcels, this quantity is an operational comparison metric rather than a direct measurement of steady-state industrial plant capacity.

The time-averaged strict accuracy is calculated by pooling all valid downstream pair observations:

$$A_{5\,\mathrm{mm}}^{\mathrm{time}} = 100\frac{\sum_{t\in\mathcal{T}_{\mathrm{valid}}}\sum_{k=1}^{K_t-1}\mathbb{I}\left(g_{t,k} > 5\,\mathrm{mm}\right)}{\sum_{t\in\mathcal{T}_{\mathrm{valid}}}\left(K_t-1\right)}, \tag{42}$$

where

$$\mathcal{T}_{\mathrm{valid}} = \{t : K_t \geq 2\}.$$

This is a pooled pairwise quantity, rather than an unweighted average of the accuracy computed independently at each timestep. It is retained as a trajectory diagnostic but is not used as the primary declaration of final singulation success.

Centroid-based separation quantities are also exported by the MATLAB environment for diagnostic comparison. They are not used for policy selection, because they do not account for parcel dimensions and can therefore classify overlapping footprints as separated.

For an evaluation set containing $M$ episodes, the primary accuracy is summarized through

$$\overline{A}_{5\,\mathrm{mm}} = \frac{1}{M}\sum_{e=1}^{M} A_{5\,\mathrm{mm}}^{(e)}, \tag{43}$$

$$\sigma_A = \sqrt{\frac{1}{M-1}\sum_{e=1}^{M}\left(A_{5\,\mathrm{mm}}^{(e)} - \overline{A}_{5\,\mathrm{mm}}\right)^2}, \tag{44}$$

$$A_{\mathrm{worst}} = \min_{1\leq e\leq M} A_{5\,\mathrm{mm}}^{(e)}. \tag{45}$$

The mean describes average nominal performance, the standard deviation quantifies episode-to-episode dispersion in percentage points, and the worst episode exposes failure behaviour that may be hidden by the mean.

When multiple independently trained agents are considered, within-agent episode variability and variability across training seeds are reported separately. The corresponding aggregation protocol is introduced in Section 6.

# 4 PPO Configuration and Experimental Method

This section describes the PPO implementation and the experimental protocol used to develop and validate the parcel-singulation policy. Only the algorithmic concepts required to interpret the implementation are introduced. The purpose is not to provide a general review of policy-gradient methods, but to document how the actor and critic were constructed, which PPO hyperparameters were optimized, and how configuration selection, training-seed sensitivity, and non-nominal environmental robustness were assessed.

The experimental workflow is shown in Figure 5. Environment validation preceded all full training campaigns. A reference PPO configuration was then defined and used as the initial point of the high-fidelity Bayesian Optimization campaign. The optimization restricted the search to a small set of promising configurations, which were subsequently retrained and compared under a common training and evaluation budget. The retained nominal configuration was then retrained with independent random seeds, while its non-nominal robustness was assessed in a separate evaluation-only stress campaign using the saved agents.

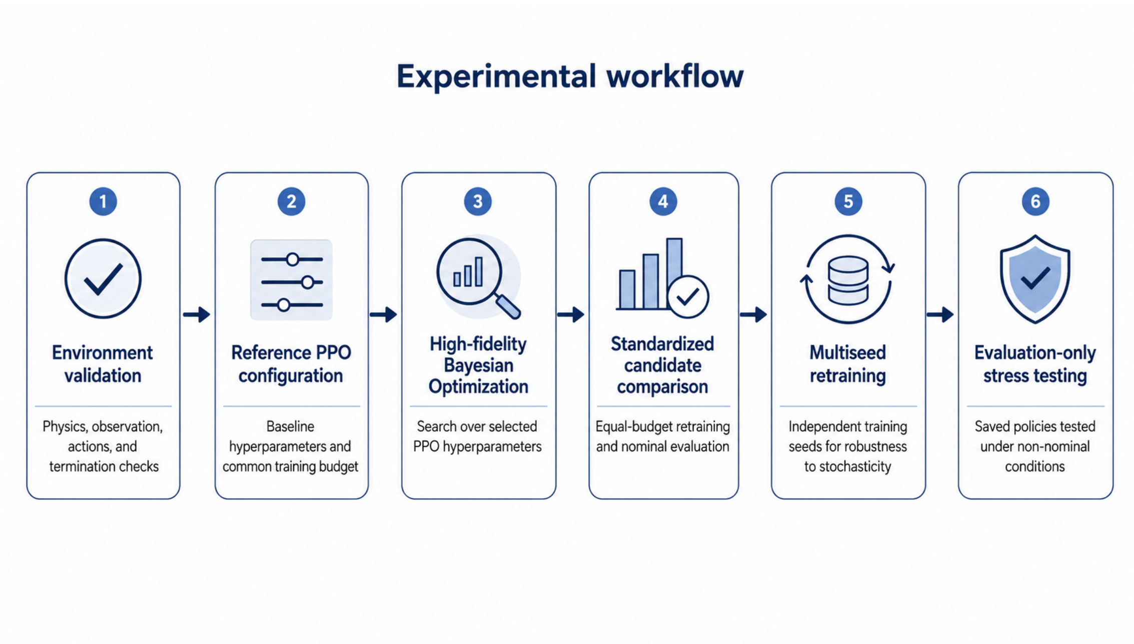


**Figure 5:** Experimental workflow used for policy development and validation. The reference PPO configuration is included as the initial point of the Bayesian Optimization campaign. Promising configurations identified by the search are retrained and compared under a common training and evaluation budget. Multiseed retraining then assesses sensitivity to training stochasticity, while the final stress campaign evaluates the saved policies without additional training.

## 4.1 PPO agent and network architecture

Proximal Policy Optimization is an on-policy actor–critic algorithm. The actor represents a parameterized stochastic policy

$$\pi_\theta(\boldsymbol{a}_t \mid \boldsymbol{o}_t),$$

whereas the critic approximates the expected discounted return from the current observation through

$$V_\phi(\boldsymbol{o}_t).$$

The actor parameters are denoted by $\theta$, while $\phi$ denotes the critic parameters.

PPO updates the actor using data generated by the current policy. Defining the probability ratio

$$\rho_t(\theta) = \frac{\pi_\theta(\boldsymbol{a}_t \mid \boldsymbol{o}_t)}{\pi_{\theta_{\text{old}}}(\boldsymbol{a}_t \mid \boldsymbol{o}_t)}, \tag{46}$$

the clipped surrogate objective is

$$\mathcal{L}^{\text{clip}}(\theta) = \mathbb{E}_t\left[\min\left(\rho_t(\theta)\widehat{A}_t, \text{clip}\left(\rho_t(\theta), 1-\epsilon, 1+\epsilon\right)\widehat{A}_t\right)\right], \tag{47}$$

where $\epsilon$ is the PPO clipping parameter and $\widehat{A}_t$ is an estimate of the advantage associated with the sampled action [2], [3]. The clipping operation limits the benefit of excessively large policy changes while still allowing several optimization epochs over the collected on-policy experience.

The advantage is estimated using Generalized Advantage Estimation (GAE):

$$\delta_t = r_t + \gamma V_\phi(\boldsymbol{o}_{t+1}) - V_\phi(\boldsymbol{o}_t), \tag{48}$$

$$\widehat{A}_t = \sum_{l=0}^{H-t-1} (\gamma\lambda)^l \delta_{t+l}, \tag{49}$$

where $\gamma$ is the discount factor, $\lambda$ is the GAE factor, and $H$ denotes the available experience horizon. Lower values of $\lambda$ generally reduce variance at the cost of a more strongly bootstrapped estimate, whereas values closer to one rely more heavily on sampled future rewards.

The implemented actor is a continuous Gaussian policy. Given the 100-dimensional observation, the network produces two 50-dimensional vectors:

$$\boldsymbol{\mu}_\theta(\boldsymbol{o}_t) \in [-1, 1]^{50}, \qquad \boldsymbol{\sigma}_\theta(\boldsymbol{o}_t) \in \mathbb{R}^{50}_{>0}. \tag{50}$$

The first vector defines the action-distribution mean, while the second defines a positive standard deviation. The sampled normalized action is subsequently limited to the action interval described in Section 2.4.

Actor and critic are implemented as separate feed-forward networks and do not share trainable features. Their architectures are summarized in Table 3.

**Table 3:** Actor and critic network architectures used in all reported PPO configurations.

| Network | Input and shared layers | Output branch | Interpretation |
|---|---|---|---|
| Actor mean | $100 \to 512 \to 256$, ReLU activations | $256 \to 128 \to 50$, ReLU then tanh | Mean of the Gaussian policy in the normalized action space. |
| Actor standard deviation | Shares the actor's first two hidden layers | $256 \to 128 \to 50$, ReLU and softplus output | Positive standard deviation of the Gaussian policy. |
| Critic | $100 \to 512 \to 256 \to 128$, ReLU activations | $128 \to 1$, linear output | Approximation of the state-value function $V_\phi(\boldsymbol{o}_t)$. |

No additional observation-normalization layer is included in either network. Numerical scaling is instead provided by the normalized grid-centric observation defined in Section 2.3. The network architecture was kept fixed during Bayesian Optimization so that the observed differences could be attributed to the selected PPO hyperparameters rather than to simultaneous changes in representation capacity.

## 4.2 Reference PPO configuration

The reference PPO configuration defines the common starting point of the hyperparameter-selection study. It is the configuration explicitly supplied as the initial point of the Bayesian Optimization campaign and is trained and evaluated using the same computational budget adopted for the shortlisted optimized candidates.

It is therefore the sole PPO baseline used in the nominal comparison presented in this report. The configuration is project-specific and originates from the supplied MATLAB implementation and the course reference settings. It is treated as a reproducible comparison point rather than as an assumed optimum. Its hyperparameters are reported in Table 4.

**Table 4:** Reference PPO hyperparameters and training settings.

| Parameter | Reference value | Role in PPO training |
|---|---:|---|
| `ExperienceHorizon` | 500 | Number of interaction steps collected before a PPO update. |
| `ClipFactor` | 0.20 | Defines the probability-ratio interval $[1-\epsilon, 1+\epsilon]$. |
| Actor learning rate | $1.0 \times 10^{-4}$ | Step size for the policy-network optimizer. |
| Critic learning rate | $1.0 \times 10^{-4}$ | Step size for the value-network optimizer. |
| `GAEFactor` | 0.95 | GAE parameter $\lambda$. |
| `DiscountFactor` | 0.99 | Discount factor $\gamma$ applied to future rewards. |
| `EntropyLossWeight` | 0.01 | Weight of the entropy term promoting policy exploration. |
| `NumEpoch` | 3 | Number of optimization passes over each collected experience set. |
| Agent sample time | 0.01 s | Equal to the environment integration timestep. |
| Training episodes | 2000 | Common training budget used in the nominal comparison. |
| Maximum steps per episode | 1000 | Safety limit applied independently of physical termination. |
| Training seed | 1 | Common initialization and training seed for nominal comparison. |
| Reward averaging window | 100 episodes | Window used only for the displayed training diagnostic. |

Both the reference configuration and the shortlisted Bayesian-Optimization candidates are trained for the complete budget of 2000 episodes. Training is not terminated when the displayed average reward reaches a prescribed value. A custom logging callback records the training progression but always returns a false stopping condition. This ensures that every configuration receives the same interaction budget independently of its apparent learning speed.

The `ScoreAveragingWindowLength` affects the displayed `AverageReward` curve but not the physical objective or the PPO update. Intermediate logging is performed every 50 episodes. These logging events are not intermediate policy checkpoints: the trained agent is saved after the training run, together with its configuration, training statistics, training curve, and post-training evaluation metrics.

Only the actor and critic learning rates are explicitly set through `rlOptimizerOptions`. Other optimizer properties, including the mini-batch configuration, remain at the defaults of the installed MATLAB Reinforcement Learning Toolbox. Optimizer properties not explicitly overridden retain the defaults of the MATLAB Reinforcement Learning Toolbox installation used for the experiments [4].

### 4.3 Nominal evaluation protocol

Each trained nominal agent is evaluated over 100 complete episodes in `RL_environment_v12`. The maximum evaluation length is 1000 steps, although normally completed episodes terminate earlier through the physical front-edge condition introduced in Section 2.5.

For episode $e$, the random seed is assigned as

$$s_e = s_{\text{base}} + e, \qquad e = 1, \dots, 100. \tag{51}$$

The standardized reference and candidate comparison uses

$$s_{\text{base}} = 3000,$$

and therefore evaluates every configuration on the same seeds $3001, \dots, 3100$.

The seed is set before creating and resetting the environment. The saved agent is then queried at every timestep through MATLAB's `getAction` interface. No separate deterministic mean-policy evaluator is defined in the project scripts. Setting the seed before the complete rollout therefore makes both the environment randomization and any stochasticity introduced by the action selection reproducible.

Using common evaluation seeds is important because it pairs the compared configurations with the same sequence of nominal test scenarios. Differences between their aggregated metrics are consequently less affected by variation in parcel sizes and initial positions.

For each episode, the evaluator records:

- whether the physical terminal condition was reached;
- the episode duration and number of steps;
- the complete reward history;
- final strict and permissive edge-gap accuracies;
- final mean and minimum gaps and maximum overlap depth;
- time-averaged downstream metrics;
- equivalent throughput;
- collision-correction count.

The primary comparison quantity is

$$\overline{A}_{5\,\text{mm}} = \frac{1}{100} \sum_{e=1}^{100} A_{5\,\text{mm}}^{(e)}.$$

Standard deviation, worst episode, overlap, throughput, and collision corrections are retained as complementary evidence. The training return is reported only as a learning diagnostic and is not included in the physical success criterion.

## 4.4 Bayesian Optimization protocol

Bayesian Optimization is used as an outer loop around complete PPO training and evaluation runs. A point in the search space represents a PPO hyperparameter configuration

$$\boldsymbol{\xi} = [\epsilon,\, \alpha_{\text{actor}},\, \alpha_{\text{critic}},\, c_{\mathcal{H}},\, \lambda,\, \gamma,\, N_{\text{epoch}},\, H_{\text{exp}}]\,,$$

where $c_{\mathcal{H}}$ denotes the entropy-loss weight and $H_{\text{exp}}$ the experience horizon.

For every objective evaluation, a new PPO agent is initialized and trained for 2000 episodes using training seed 1. It is subsequently evaluated on the common set of 100 nominal episodes with evaluation seed base 3000. The Bayesian Optimization objective is

$$J_{\text{BO}}(\boldsymbol{\xi}) = -\overline{A}_{5\,\text{mm}}(\boldsymbol{\xi}), \tag{52}$$

because MATLAB's `bayesopt` routine performs minimization. Neither final training reward nor `AverageReward` enters the optimization objective.

The high-fidelity search uses 20 objective evaluations. The reference PPO configuration reported in Table 4 is explicitly supplied as the initial point of the optimization. Subsequent configurations are selected through the `expected-improvement-plus` acquisition function. Fixed training and evaluation seeds are used for all objective evaluations, and the objective is declared deterministic to `bayesopt`. This choice reduces scenario-induced noise during hyperparameter screening, while sensitivity to PPO training stochasticity is examined separately through the later multiseed campaign.

The search space is reported in Table 5.

**Table 5:** PPO hyperparameter search space used in the high-fidelity Bayesian Optimization campaign.

| Hyperparameter | Search domain | Sampling type |
|---|---|---|
| `ClipFactor` | $[0.05,\, 0.30]$ | Continuous linear scale |
| Actor learning rate | $[3 \times 10^{-5},\, 3 \times 10^{-4}]$ | Continuous logarithmic scale |
| Critic learning rate | $[3 \times 10^{-5},\, 3 \times 10^{-4}]$ | Continuous logarithmic scale |
| `EntropyLossWeight` | $[10^{-4},\, 2 \times 10^{-2}]$ | Continuous logarithmic scale |
| `GAEFactor` | $[0.90,\, 0.99]$ | Continuous linear scale |
| `DiscountFactor` | $[0.970,\, 0.995]$ | Continuous linear scale |
| `NumEpoch` | $\{2, 3, 4, 5\}$ | Integer |
| `ExperienceHorizon` | $\{400, 500, 750, 1000\}$ | Categorical |

Throughout the search, the following components are held fixed:

- the nominal physical environment;
- parcel generation and termination logic;
- observation and action definitions;
- action denormalization;
- training reward;
- actor and critic architectures;
- training and evaluation budgets;

- training and evaluation seed sets;
- final physical metric implementation.

The Bayesian Optimization campaign is used to restrict the original search space to a small set of promising PPO configurations. Since the trained agents generated during the optimization campaign were not retained, the shortlisted configurations are subsequently retrained from scratch. The same procedure is also applied to the reference PPO configuration, so that all retained configurations are compared under an equal computational and evaluation budget.

Each shortlisted configuration is trained for 2000 episodes using the common nominal training seed and is evaluated over the same 100 nominal episodes. This standardized comparison serves two purposes: it produces the saved agents required for subsequent policy analysis and provides a controlled final validation of the configurations proposed by the Bayesian Optimization search.

Because PPO training is stochastic, the ordering observed during this final retraining need not reproduce exactly the ranking of individual objective evaluations within the optimization trace. The final nominal configuration is therefore selected from the standardized comparison of the reference configuration and the shortlisted Bayesian-Optimization candidates, rather than from a single internal objective evaluation.

The complete process is treated in this report as one BO-based policy-selection workflow:

- reference configuration and search space
- Bayesian Optimization
- shortlisted configurations
- standardized final comparison.

The resulting nominal comparison and the configuration retained for subsequent multiseed analysis are presented in Section 5.

### 4.5 Multiseed and stress-test protocols

The BO-based nominal selection workflow uses one common training seed to reduce configuration-comparison noise. This controlled comparison does not establish that the retained PPO configuration is insensitive to neural-network initialization, trajectory collection, and stochastic policy optimization. A separate multiseed stage is therefore performed after nominal policy selection.

The selected hyperparameter configuration is retrained independently using

$$\mathcal{S}_{\text{train}} = \{1,\ 278468,\ 287102,\ 270632\}\,. \tag{53}$$

Each run uses 2000 training episodes and produces a separately saved agent. All four agents are then evaluated on the same 100 nominal episodes using evaluation seed base 5000. This protocol separates:

- within-agent variability across evaluation episodes;
- between-agent variability caused by the PPO training seed.

The multiseed stage changes only the training seed. Environment, reward, observation, action definition, network architecture, PPO hyperparameters, training budget, and physical metrics remain unchanged.

After completing the multiseed validation, the four saved policies are loaded for the non-nominal stress campaign. This stage is strictly *evaluation-only*:

- no policy parameters are updated;
- no additional training episodes are collected;
- the PPO hyperparameters are not changed;
- the reward is not redesigned;
- each policy is tested under the same non-nominal case definitions.

The standard stress campaign uses 100 episodes per case and per saved agent, with evaluation seed base 2044876 and a maximum of 2000 steps per episode. One environmental factor is modified at a time in the standard campaign. A separate feasible three-parcel inlet case is evaluated through the dedicated batch-cardinality environment. The exact stress definitions and corresponding results are reported in Section 6.

This final distinction is central to the interpretation of the study: multiseed validation measures whether the same PPO configuration can be learned reliably, whereas the stress campaign measures whether already learned policies generalize when the operating conditions differ from the nominal training distribution.

# 5 Nominal Policy Development and Selection

This section presents the nominal results of the BO-based policy-selection workflow introduced in Section 4. The purpose is to compare the reference PPO configuration and the shortlisted optimized configurations under the same training and evaluation conditions, and to identify the nominal policy to be carried forward to the multiseed and non-nominal analyses.

All configurations discussed in this section use the same physical environment, observation, action space, reward, actor–critic architecture, training budget, and nominal evaluation episodes. Consequently, the comparison isolates the effect of the selected PPO hyperparameters.

## 5.1 Reference PPO performance

The reference PPO configuration defined in Section 4.2 constitutes the sole baseline for the nominal comparison. It was trained for 2000 episodes using training seed 1 and subsequently evaluated over 100 nominal episodes using evaluation seed base 3000.

Its physical and operational results are reported in Table 6.

**Table 6:** Standardized nominal validation of the reference PPO configuration.

| Metric | Value |
|---|---|
| Mean final edge-gap accuracy, $g > 5\,\mathrm{mm}$ | 93.44 % |
| Episode-to-episode standard deviation | 8.66 pp |
| Worst episode accuracy | 66.67 % |
| Mean maximum overlap depth | 0.0500 m |
| Mean equivalent throughput | 8250.1 parcels/h |
| Mean collision-correction count | 20.53 |
| Non-completion rate | 0 % |

The reference policy already achieves a high mean strict accuracy and completes every nominal evaluation episode. It therefore provides evidence that the validated observation, reward, and PPO implementation are capable of learning a physically meaningful singulation strategy.

The mean performance alone, however, does not establish consistent behaviour. The standard deviation of 8.66 percentage points and the worst episode of 66.67 % show that a substantial fraction of parcel pairs can remain insufficiently separated in unfavourable nominal realizations. The mean maximum overlap depth of 0.0500 m and the mean collision-correction count of 20.53 provide additional evidence that the policy still produces physically undesirable trajectories in some episodes.

These results define the improvement target for PPO hyperparameter tuning: increase final edge-based accuracy while reducing dispersion, residual overlap, and dependence on numerical collision correction, without compromising episode completion or operational throughput.

## 5.2 Shortlisted Bayesian-Optimization configurations

The high-fidelity Bayesian Optimization campaign identified a restricted set of promising PPO configurations. As described in Section 4.4, the agents generated internally during the optimization campaign were not retained. The shortlisted configurations were therefore retrained from scratch and evaluated through the common standardized protocol.

The three retained configurations are denoted `BO_008`, `BO_018`, and `BO_014`. These identifiers label the conceptual hyperparameter sets used in the final comparison and should not be interpreted as the final ranking of the configurations.

Their settings are reported together with the reference PPO configuration in Table 7.

**Table 7:** PPO hyperparameters used in the standardized nominal candidate comparison.

| Configuration | Clip | Actor LR | Critic LR | Entropy | GAE | Discount | Epochs | Horizon |
|---|---|---|---|---|---|---|---|---|
| Reference PPO | 0.200 | $1.00 \times 10^{-4}$ | $1.00 \times 10^{-4}$ | $1.00 \times 10^{-2}$ | 0.950 | 0.990 | 3 | 500 |
| `BO_008` | 0.201 | $4.78 \times 10^{-5}$ | $2.94 \times 10^{-4}$ | $8.99 \times 10^{-3}$ | 0.961 | 0.993 | 4 | 400 |
| `BO_018` | 0.097 | $6.37 \times 10^{-5}$ | $8.88 \times 10^{-5}$ | $4.31 \times 10^{-3}$ | 0.986 | 0.992 | 4 | 400 |
| `BO_014` | 0.098 | $5.66 \times 10^{-5}$ | $2.96 \times 10^{-4}$ | $1.60 \times 10^{-4}$ | 0.959 | 0.995 | 4 | 400 |

The optimized configurations do not correspond to a uniform increase or decrease of all PPO parameters. Nevertheless, several recurring tendencies can be identified.

All shortlisted configurations use an experience horizon of 400 steps and four optimization epochs, compared with 500 steps and three epochs for the reference configuration. The actor learning rates are consistently lower than the reference value. By contrast, the critic learning rate is configuration dependent: it is strongly increased in `BO_008` and `BO_014`, but remains close to the reference order of magnitude in `BO_018`.

The two strongest final candidates, `BO_018` and `BO_014`, also use a substantially smaller clip factor than the reference configuration. Their entropy-loss weights are reduced, particularly for `BO_014`. These observations describe the configurations retained by the search, but they do not establish the isolated causal effect of each hyperparameter, because the parameters were optimized jointly.

### 5.3 Interpretation of the hyperparameter shifts

The shortlisted configurations reveal a change in the balance between policy-update conservativeness, value-function adaptation, exploration, and reuse of recently collected on-policy experience. These changes should be interpreted as coupled design choices rather than as independent causal effects.

First, both `BO_018` and `BO_014` reduce the PPO `ClipFactor` from 0.20 to approximately 0.10. The probability ratio introduced in Equation 46 is therefore constrained within a narrower interval:

$$\text{Reference:} \qquad \rho_t \in [0.80,\, 1.20], \tag{54}$$

$$\texttt{BO\_014}: \qquad \rho_t \in [0.902,\, 1.098]. \tag{55}$$

This tighter clipping range limits the contribution of large changes in action probability during an individual PPO update. In the present 50-dimensional continuous-action problem, this may reduce destructive policy changes after a batch of experience has been collected.

The actor learning rate is also reduced in every shortlisted configuration. For `BO_014`, it decreases from

$$1.00 \times 10^{-4} \quad \text{to} \quad 5.66 \times 10^{-5}.$$

The lower actor learning rate and smaller clip factor both act in the direction of more conservative policy adaptation. This may be beneficial because the singulation strategy depends on coordinated outputs across several AMS cells: an abrupt update affecting many action components simultaneously could disrupt a previously useful coordination pattern.

At the same time, all shortlisted candidates reduce the `ExperienceHorizon` from 500 to 400 and increase `NumEpoch` from 3 to 4. PPO updates are therefore triggered after a smaller quantity of newly collected experience, while each experience set is reused for one additional optimization pass. The resulting regime can be interpreted as more frequent updates with slightly greater reuse of recent on-policy samples.

This does not necessarily make actor learning more aggressive. In `BO_014`, the additional epoch and shorter horizon are counterbalanced by the smaller actor learning rate and tighter clipping. The combined effect is more consistent with frequent but controlled policy refinement than with larger policy changes.

The critic learning rate follows a different pattern. In `BO_014`, it increases from

$$1.00 \times 10^{-4} \quad \text{to} \quad 2.96 \times 10^{-4},$$

while the actor learning rate is nearly halved. The final configuration therefore creates a marked asymmetry between actor and critic adaptation.

A more responsive critic may track the returns generated by the current policy more rapidly. Since the critic supplies the value estimates used in the GAE advantage calculation, improved fitting of the value function may reduce error and variance in the advantage estimates supplied to the actor. The policy can then evolve cautiously while being supported by a critic that adapts more quickly to the current return distribution.

This is a plausible actor–critic interpretation, not a directly isolated experimental result. No critic-loss ablation or value-prediction validation was performed to demonstrate that the increased critic learning rate alone caused the observed improvement.

The entropy-loss weight is strongly reduced in the final configuration:

$$1.00 \times 10^{-2} \quad \longrightarrow \quad 1.60 \times 10^{-4}.$$

The reference setting applies a substantially stronger pressure to preserve entropy in the Gaussian action distribution. The lower optimized value permits the policy to become more concentrated around preferred AMS commands once a useful singulation behaviour has emerged.

This may improve repeatability in a task where consistent actuator coordination is more important than maintaining broad exploration indefinitely. The corresponding risk is premature loss of exploration, which could increase sensitivity to initialization or cause convergence to different local strategies. The multiseed analysis is therefore necessary to determine whether the final configuration remains reliable under different training realizations.

The discount factor increases slightly, from 0.990 to approximately 0.995. This change is consistent with the delayed structure of the task: the consequences of an action applied on the AMS become measurable only after the parcel enters the downstream reward region. A larger discount factor reduces the attenuation of these delayed downstream outcomes.

For example, a reward received $n$ steps after an action is weighted by $\gamma^n$. Since the environment timestep is 0.01 s, even a small increase in $\gamma$ can noticeably alter the relative weight of effects observed several hundred timesteps later. Nevertheless, because discount factor, GAE factor, actor–critic learning rates, and reward structure interact, this interpretation should not be treated as an isolated explanation of the final performance.

The GAE factor changes only moderately. Its final value of approximately 0.959 remains close to the reference value of 0.950. The selected configuration therefore retains a broadly similar bias–variance trade-off in advantage estimation. The more pronounced changes concern:

- the smaller clip factor;
- the lower actor learning rate;
- the higher critic learning rate;
- the much lower entropy-loss weight;
- the shorter experience horizon;
- the additional optimization epoch.

Overall, `BO_014` can be interpreted as combining a more conservative actor with a more responsive critic:

$$\underbrace{\text{smaller clip factor} + \text{lower actor learning rate}}_{\text{controlled policy adaptation}} \quad + \quad \underbrace{\text{higher critic learning rate} + \text{one additional epoch}}_{\text{faster value-function adaptation}}. \tag{56}$$

The strongly reduced entropy weight allows the policy to exploit a stable actuation strategy, while the slightly larger discount factor preserves the importance of delayed downstream separation.

This combined interpretation is compatible with the lower nominal dispersion and residual overlap observed for `BO_014`. However, the experiment does not identify the independent contribution of each parameter. A controlled ablation study would be required to establish causal attribution.

## 5.4 Standardized nominal comparison

The reference configuration and all shortlisted candidates were trained for the complete 2000-episode budget and evaluated on the same 100 nominal episodes. Table 8 reports the resulting physical and operational metrics.

**Table 8:** Standardized 2000-episode nominal validation of the reference PPO configuration and the selected Bayesian-Optimization candidates.

| Configuration | Mean acc. > 5 mm [%] | Std [pp] | Worst [%] | Mean max overlap [m] | Throughput [pph] | Collisions | Non-completion [%] |
|---|---|---|---|---|---|---|---|
| Reference PPO | 93.44 | 8.66 | 66.67 | 0.0500 | 8250.1 | 20.53 | 0 |
| `BO_008` | 97.33 | 4.77 | 88.89 | 0.0222 | 8052.1 | 5.05 | 0 |
| `BO_018` | 98.89 | 3.35 | 88.89 | 0.00352 | 8244.7 | 0.25 | 0 |
| `BO_014` | 99.78 | 1.56 | 88.89 | 0.00133 | 7874.2 | 1.00 | 0 |

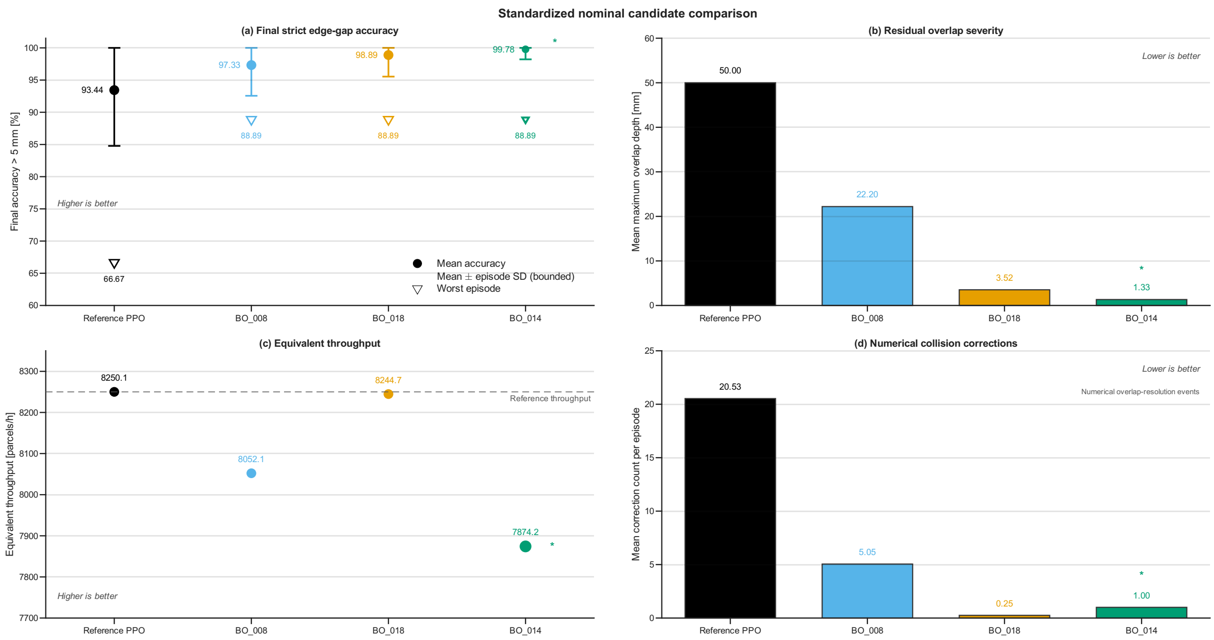


**Figure 6:** Standardized nominal comparison of the reference PPO configuration and the three shortlisted Bayesian-Optimization candidates. The figure reports the mean strict edge-gap accuracy with episode-level dispersion together with the worst episode, residual overlap, equivalent throughput, and collision-correction diagnostics.

All three BO-tuned configurations improve the primary physical metric relative to the reference PPO configuration. The improvement is not limited to the mean accuracy. Each candidate also reduces the episode-to-episode standard deviation, increases the worst evaluated accuracy from 66.67 % to 88.89 %, and decreases the mean maximum overlap depth.

The improvement from the reference configuration to `BO_008` is already substantial:

$$93.44\,\% \longrightarrow 97.33\,\%$$

in mean strict accuracy, accompanied by an approximate halving of the standard deviation and a reduction of the mean maximum overlap from 50.0 mm to 22.2 mm. The collision-correction count also falls from 20.53 to 5.05.

The two strongest candidates, `BO_018` and `BO_014`, further reduce the remaining overlap. Their mean maximum overlap depths are only 3.52 mm and 1.33 mm, respectively. This is particularly relevant because a high percentage accuracy could otherwise coexist with a small number of severe failure cases. Here, the overlap diagnostic confirms that the residual failures are also less physically severe on average than in the reference policy.

The operational metrics reveal a trade-off. `BO_018` preserves almost the same throughput as the reference PPO configuration:

$$8244.7 \text{ pph} \quad \text{versus} \quad 8250.1 \text{ pph},$$

while producing the smallest collision-correction count of the four evaluated policies. `BO_014`, in contrast, achieves the strongest singulation quality but has the lowest mean throughput, 7874.2 parcels/h.

No configuration produces non-completion within the evaluation horizon. Therefore, the reported differences concern singulation quality and completion time rather than the ability to finish the nominal task.

The results also confirm why the training return cannot be used as the sole selection criterion. The final ranking is determined by independently computed edge-based and operational quantities, rather than by the numerical scale of the accumulated training reward.

## 5.5 Final nominal selection

The final nominal configuration selected for the subsequent analyses is `BO_014`. Its complete PPO parameter set is

$$\texttt{ExperienceHorizon} = 400, \tag{57}$$

$$\texttt{ClipFactor} = 0.0983614593, \tag{58}$$

$$\alpha_{\text{actor}} = 5.66136285 \times 10^{-5}, \tag{59}$$

$$\alpha_{\text{critic}} = 2.96096801 \times 10^{-4}, \tag{60}$$

$$\lambda = 0.959291505, \tag{61}$$

$$\gamma = 0.994780672, \tag{62}$$

$$\texttt{EntropyLossWeight} = 1.59779195 \times 10^{-4}, \tag{63}$$

$$\texttt{NumEpoch} = 4. \tag{64}$$

The selection is based primarily on three pieces of evidence:

- the highest mean final strict accuracy, 99.78 %;
- the smallest episode-to-episode standard deviation, 1.56 percentage points;
- the smallest mean maximum overlap depth, 1.33 mm.

The worst evaluated episode is 88.89 %, equal to those of the other shortlisted candidates and markedly higher than the reference result. Since a nominal episode contains nine evaluated gaps, this value corresponds to one unsuccessful consecutive pair in the worst observed episode.

The selection does not imply that `BO_014` dominates every operational metric. Its mean throughput is approximately 4.6% lower than that of the reference configuration and `BO_018`. Nevertheless, it remains above the nominal 6000 parcels/h requirement discussed in the methodological reference [1]. The throughput reduction is therefore accepted in exchange for higher final separation accuracy, lower variability, and smaller residual overlap.

`BO_018` remains a relevant operational alternative. It combines 98.89 % mean strict accuracy with reference-like throughput and the lowest collision-correction count. It would be a reasonable choice if throughput or reduced numerical contact correction were prioritized more strongly. The objective of the present project, however, is the physical quality and consistency of longitudinal singulation. On that basis, `BO_014` is retained.

This conclusion is initially based on one standardized training realization. It establishes which saved nominal policy performs best among the evaluated candidates, but it does not yet demonstrate that the corresponding hyperparameter configuration can be learned consistently under different training seeds. That question is addressed through the independent multiseed retraining campaign presented in Section 6.

# 6 Multiseed Validation and Non-Nominal Robustness

The standardized nominal comparison identifies the strongest saved policy among the evaluated PPO configurations, but it does not by itself establish that the corresponding hyperparameters are reliable under repeated training or that the learned behaviour generalizes beyond the nominal environment.

Two complementary robustness questions are therefore considered in this section:

1. whether the selected PPO configuration can be learned consistently from different random training realizations;
2. whether the resulting saved policies remain effective when the physical and operational conditions differ from those used during training.

The first question is addressed through independent multiseed retraining. The second is addressed through an evaluation-only stress campaign. These analyses must not be conflated: multiseed variability concerns the stochastic optimization process, whereas stress-test degradation concerns distribution shift in the controlled environment.

## 6.1 Multiseed validation protocol

The final `BO_014` hyperparameter configuration selected in Section 5.5 was retrained independently using the four training seeds

$$\mathcal{S}_{\text{train}} = \{1,\, 278468,\, 287102,\, 270632\}\,. \tag{65}$$

Every run used the same:

- nominal environment;
- observation and action definitions;
- reward function;
- actor and critic architectures;
- PPO hyperparameters;
- 2000-episode training budget;
- maximum episode length.

Only the random seed controlling network initialization, environment randomization, trajectory generation, and stochastic PPO training was changed. Each completed training produced a separate saved policy.

The four agents were subsequently evaluated over 100 nominal episodes each, using evaluation seed base 5000. The same episode seeds were used for every agent. This common test set reduces variability due to parcel dimensions and inlet positions and makes differences between policies primarily attributable to their independent training realizations.

For agent $s$, the mean nominal strict accuracy is

$$\overline{A}_s = \frac{1}{100} \sum_{e=1}^{100} A_{5\,\mathrm{mm}}^{(s,e)}. \tag{66}$$

The multiseed mean and the between-training-seed standard deviation are then defined as

$$\overline{A}_{\mathrm{MS}} = \frac{1}{4} \sum_{s \in \mathcal{S}_{\mathrm{train}}} \overline{A}_s, \tag{67}$$

$$\sigma_{\mathrm{train}} = \sqrt{\frac{1}{3} \sum_{s \in \mathcal{S}_{\mathrm{train}}} \left(\overline{A}_s - \overline{A}_{\mathrm{MS}}\right)^2}. \tag{68}$$

The quantity $\sigma_{\mathrm{train}}$ must be distinguished from the episode-level standard deviation calculated separately for each trained agent. The former measures differences between the four PPO training outcomes; the latter measures variability across nominal test episodes for one fixed policy.

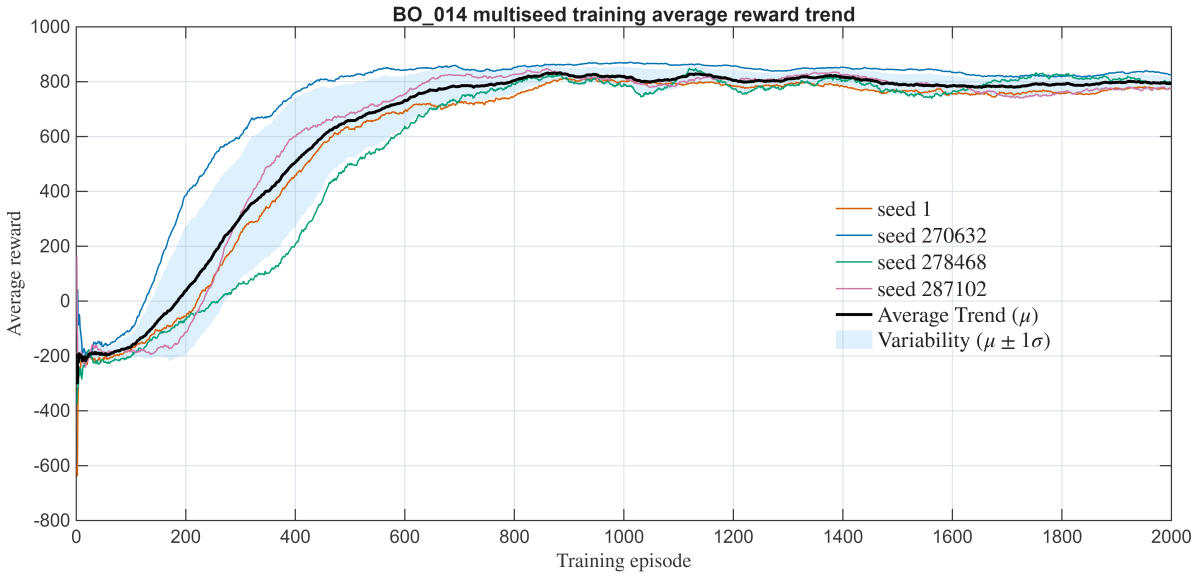


**Figure 7:** Average-reward trajectories obtained from the four independent 2000-episode training runs of the selected `BO_014` configuration. The coloured curves correspond to the individual training seeds, while the black curve and shaded region represent the cross-seed mean and one cross-seed standard deviation, respectively. The figure is used only as a training diagnostic; physical singulation performance is evaluated independently through the edge-based metrics reported in Table 9.

The training trajectories exhibit a clear dependence on the random seed during the transient learning phase. In particular, the agents reach the high-reward region at different rates, producing substantial cross-seed variability during approximately the first half of training. The variability band subsequently contracts as all four trajectories approach a broadly comparable reward plateau.

No training run exhibits a persistent collapse or a failure to improve. Nevertheless, the different convergence rates confirm that the stochastic PPO training process does not produce identical learning histories. The reward curves are therefore useful for identifying gross instabilities and differences in learning speed, but they do not determine which trained policy achieves the best physical singulation.

The quantitative multiseed evaluation is consequently reported separately in Table 9. As established in Section 3, the accumulated training reward depends on the complete temporal evolution of downstream parcel configurations and is not equivalent to final strict edge-gap accuracy.

## 6.2 Multiseed nominal performance

Since the training reward is not a physical success metric, the nominal evaluation results are kept separate from the training trajectories. Table 9 reports the edge-based and collision-related results of the four independently trained agents, with each row summarizing 100 common nominal evaluation episodes.

**Table 9:** Nominal physical evaluation of the selected `BO_014` configuration across independent PPO training seeds. Each row summarizes 100 nominal evaluation episodes.

| Training seed | Mean accuracy $> 5\,\mathrm{mm}$ [%] | Episode SD [pp] | Worst episode [%] | Mean max overlap [m] | Mean collision corrections [episode$^{-1}$] |
|---:|---:|---:|---:|---:|---:|
| 1 | 99.67 | 1.90 | 88.89 | 0.0044 | 0.71 |
| 278468 | 96.67 | 6.01 | 77.78 | 0.0255 | 12.74 |
| 287102 | 95.33 | 6.55 | 77.78 | 0.0248 | 27.24 |
| 270632 | 99.78 | 1.56 | 88.89 | 0.0007 | 3.31 |

All four policies retain a mean strict accuracy above 95 %. The selected hyperparameter configuration therefore does not depend on one isolated successful training realization, and none of the four runs exhibits a collapse of the final singulation behaviour.

The two strongest agents, obtained with seeds 1 and 270632, achieve mean accuracies close to 100 % with low episode-to-episode dispersion. The agents trained with seeds 278468 and 287102 remain effective, but display larger variability and worst episodes of 77.78 %.

The differences are more pronounced in the secondary physical diagnostics. The mean maximum overlap varies from approximately 0.7 mm to 25.5 mm, while the mean collision-correction count ranges from 0.71 to 27.24 events per episode. In particular, the agent trained with seed 287102 achieves a mean strict accuracy of 95.33 % despite accumulating substantially more collision corrections than the agents trained with seeds 1 and 270632.

This result shows that similar final edge-based accuracies can be reached through different transient parcel trajectories. Final singulation quality is therefore comparatively robust, while dynamic congestion and reliance on numerical overlap correction remain more sensitive to the training seed.

Aggregating the four agents gives

$$\overline{A}_{\mathrm{MS}} = 97.861\,\%, \tag{69}$$

$$\sigma_{\mathrm{train}} = 2.217 \text{ percentage points.} \tag{70}$$

The best and worst per-seed mean accuracies are 99.778 % and 95.333 %, respectively, while the worst individual episode across all 400 nominal rollouts reaches 77.778 %.

The additional aggregate diagnostics are:

$$\overline{A}_{g>0} = 98.028\,\%, \tag{71}$$

$$\overline{A}_{g>5\,\mathrm{mm}} = 97.861\,\%, \tag{72}$$

$$\overline{g}_{\mathrm{min,final}} = 0.1421\,\mathrm{m}, \tag{73}$$

$$\overline{o}_{\mathrm{max,final}} = 0.0139\,\mathrm{m}, \tag{74}$$

$$\overline{A}^{\mathrm{time}}_{g>5\,\mathrm{mm}} = 94.964\,\%, \tag{75}$$

$$\overline{Q} = 8018.65 \text{ parcels/h}, \tag{76}$$

$$\overline{N}_{\mathrm{collision}} = 11.0. \tag{77}$$

The permissive and strict final accuracies differ by only 0.167 percentage points. Therefore, only a small fraction of the evaluated gaps lies in the interval

$$0 < g \leq 5\,\text{mm}.$$

The strict threshold is consequently not converting a large population of nearly separated pairs into failures. Most pairs that fail the strict criterion also fail the basic positive-clearance requirement.

Overall, the multiseed campaign confirms the final configuration in terms of its primary objective. The result should nevertheless be described as *robust final singulation performance*, rather than as identical learned behaviour across seeds.

### 6.3 Evaluation-only stress-campaign design

The four saved multiseed agents were subsequently tested under non-nominal conditions. No additional training or policy update was performed during this phase. The reward, PPO hyperparameters, actor and critic networks, normalized action space, and strict physical evaluation threshold remained unchanged.

Each saved agent was evaluated over 100 episodes for every stress case, using evaluation seed base

$$s_{\text{base}} = 2044876$$

and a maximum evaluation length of 2000 steps. The reported result for one scenario therefore combines four agent-level summaries and a total of 400 evaluation rollouts.

Most cases use the parametric environment `RL_environment_v13.m`. This environment retains the 100-dimensional observation and 50-dimensional action interfaces required by the trained policy, while allowing selected physical parameters to be modified.

For the rectangular-parcel test, the observation block becomes

$$\boldsymbol{o}_c = \begin{bmatrix} x_{\text{rel},c} & y_{\text{rel},c} & w_{\text{norm},c} & l_{\text{norm},c} \end{bmatrix}^T. \tag{78}$$

During nominal square-parcel training the final two components were equal. The rectangular case breaks this equality by sampling width and longitudinal length independently, while preserving the input dimension expected by the policy.

The evaluated cases are summarized in Table 10.

**Table 10:** Definition of the nominal reference and non-nominal evaluation-only stress cases.

| Case | Modification relative to nominal training |
|---|---|
| `nominal_v13` | Ten square parcels, two per batch, generation period 0.75 s, downstream velocity 1.9 m/s. Campaign-internal reference. |
| `parcels_20` | Twenty square parcels instead of ten, yielding 19 final consecutive gaps. |
| `generation_0p60` | Two parcels generated every 0.60 s. |
| `generation_0p50` | Two parcels generated every 0.50 s. |
| `treadmill_1p60` | Passive downstream velocity reduced to 1.60 m/s. |
| `treadmill_1p30` | Passive downstream velocity reduced to 1.30 m/s. |
| `rectangular_random` | Lateral width and longitudinal length sampled independently within the nominal dimensional bounds. |
| `batch3_12` | Twelve parcels generated as four left–centre–right batches of three; period 0.75 s; square side restricted to 0.05 m to 0.30 m. |

The `nominal_v13` result is not a copy of the standardized single-configuration result reported in Section 5. It is a campaign-internal reference obtained by evaluating all four multiseed agents in the parameterized environment with the stress-campaign seed set. Differences between the two nominal values arise from the different trained agent set, evaluation seeds, and aggregation procedure. All stress effects must therefore be assessed relative to `nominal_v13`, rather than directly relative to the 99.78 % saved-agent result of Section 5.

The first seven cases modify one principal environmental factor at a time. The `batch3_12` case requires a dedicated environment because three parcels must be initialized simultaneously across a one-metre inlet.

Three parcels with the nominal maximum side of 0.40 m cannot always be placed without overlap. The maximum side is therefore reduced to 0.30 m, and a dedicated placement routine enforces non-overlap and lateral feasibility.

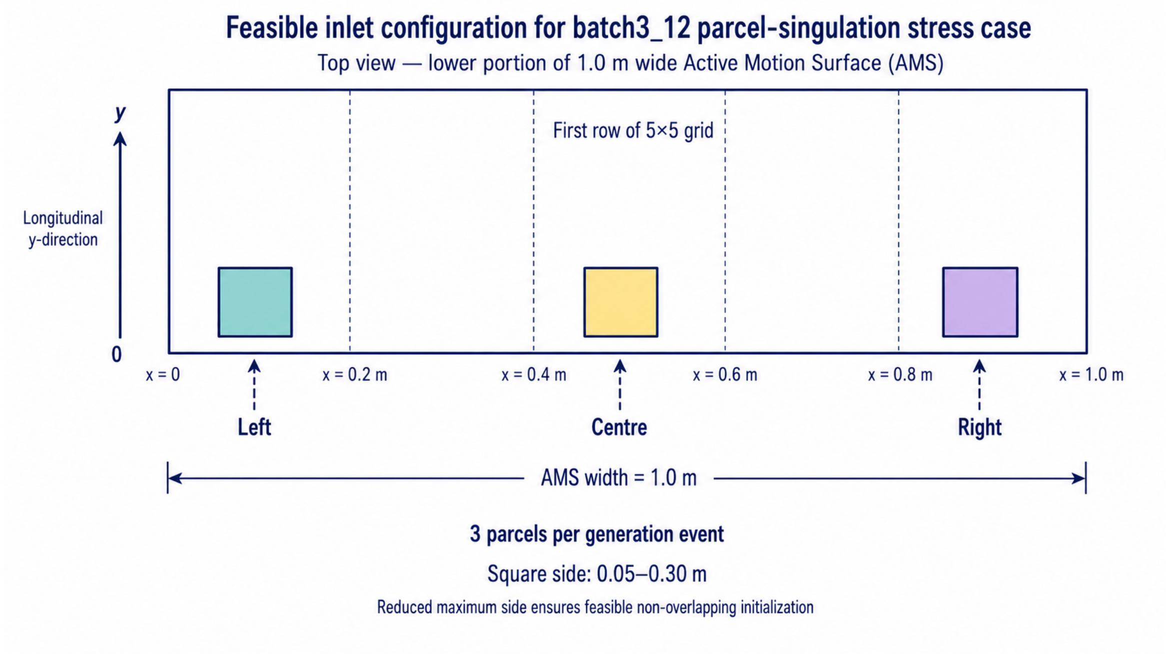


**Figure 8:** Illustrative schematic of the feasible inlet configuration used for the batch3_12 stress case. Each generation event introduces one square parcel in the left, centre, and right inlet regions. The maximum parcel side is reduced to 0.30 m so that three parcels can be initialized simultaneously inside the one-metre-wide AMS without overlap. The drawing illustrates the implemented inlet structure and does not represent one exact randomized episode initialization.

The batch3_12 test is consequently not a strict one-factor-at-a-time experiment: it changes batch cardinality and inlet placement while also restricting the parcel-size distribution for feasibility. Its result must be interpreted as robustness to a feasible three-parcel inlet structure, not as an isolated causal estimate of batch cardinality alone.

## 6.4 Stress-campaign results

Table 11 reports the aggregate results. The mean accuracy is the mean of the four agent-level mean accuracies. The associated standard deviation is the variation *across the four trained agents*, not the episode-level standard deviation within one policy.

The table also retains the worst agent-level mean as a conservative numerical diagnostic. Figure 9 instead emphasizes the cross-agent mean and dispersion, together with overlap severity and throughput, to provide a clearer visual comparison of the operating conditions.

**Table 11:** Aggregate evaluation-only stress-campaign results. Each scenario contains 100 episodes for each of the four independently trained BO_014 agents.

| Scenario | Mean acc. $> 5$ mm [%] | SD across agents [pp] | Worst agent mean [%] | Mean max overlap [m] | Throughput [pph] | Non-completion [%] |
|---|---|---|---|---|---|---|
| nominal_v13 | 98.278 | 1.825 | 96.333 | 0.00993 | 8027.3 | 0 |
| parcels_20 | 97.842 | 2.059 | 95.842 | 0.02505 | 8738.4 | 0 |
| generation_0p60 | 92.361 | 1.016 | 90.889 | 0.05790 | 9242.5 | 0 |
| generation_0p50 | 85.222 | 7.341 | 77.667 | 0.11290 | 10360.0 | 0 |
| treadmill_1p60 | 96.861 | 2.972 | 93.667 | 0.01414 | 7873.7 | 0 |
| treadmill_1p30 | 94.361 | 5.217 | 88.000 | 0.02410 | 7661.7 | 0 |
| rectangular_random | 97.417 | 2.097 | 95.000 | 0.01795 | 8035.6 | 0 |
| batch3_12 | 78.340 | 7.580 | 72.000 | 0.11280 | 11481.0 | 0 |

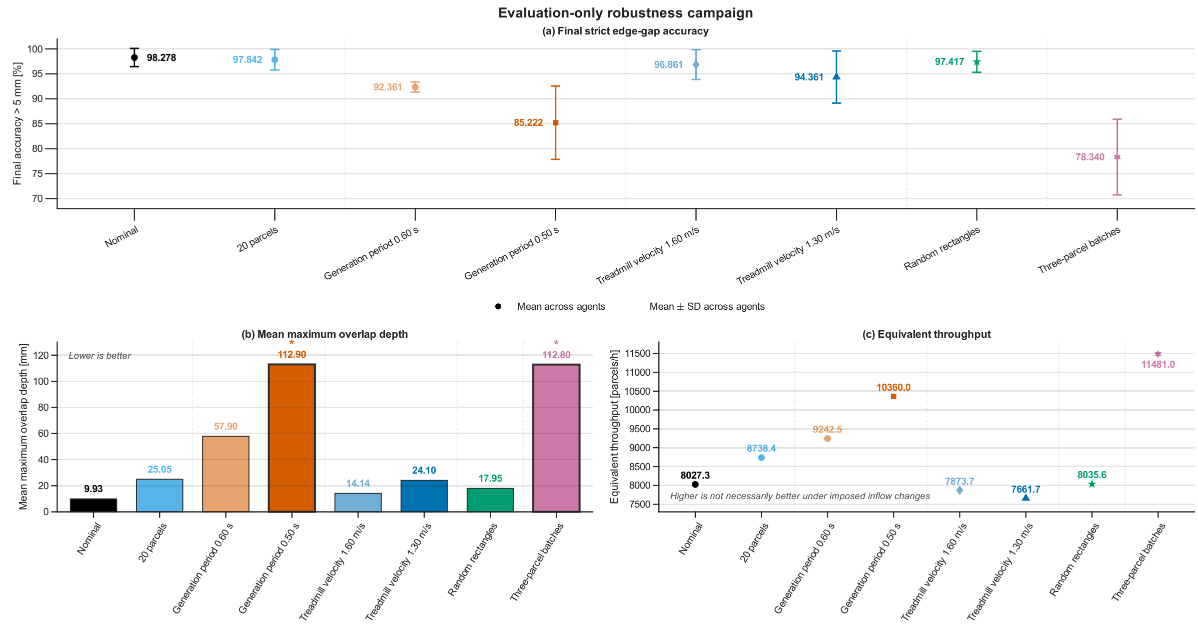


**Figure 9:** Evaluation-only stress-campaign summary. Panel (a) reports the mean final strict edge-gap accuracy across the four independently trained agents, with error bars corresponding to one standard deviation across their agent-level mean accuracies. Panels (b) and (c) report the mean maximum overlap depth and equivalent throughput, respectively. The `nominal_v13` case provides the internal reference for the campaign. The error bars represent between-agent variability and are neither confidence intervals nor episode-level standard deviations.

No tested condition produces non-completion. Every saved policy therefore remains capable of transporting all generated parcels to the final evaluation line within the prescribed evaluation horizon. This does not imply successful singulation: substantial reductions in accuracy and increases in overlap occur under the most congested inlet conditions.

## 6.5 Interpretation of robustness and operating limits

**Number of parcels.** Increasing the total parcel count from 10 to 20 produces only a minor reduction in mean strict accuracy:

$$98.278\,\% \longrightarrow 97.842\,\%.$$

The policy therefore remains effective over a longer episode containing 19 evaluated final gaps instead of 9. The mean maximum overlap increases from 9.93 mm to 25.05 mm, showing that the longer stream is more congested, but the policy does not collapse when the number of parcels is doubled.

**Parcel geometry.** The random rectangular-parcel case retains a mean strict accuracy of 97.417 %. The policy was trained with square parcels, for which the two normalized size channels were equal. The result therefore shows good generalization when width and longitudinal length become independent within the tested dimensional limits.

This conclusion applies only to the geometric range represented by the stress environment. It does not establish robustness to arbitrary aspect ratios, rotated parcels, or parcel dynamics absent from the kinematic model.

**Generation period.** Increasing the temporal density of the inlet is substantially more damaging. Reducing the generation period from 0.75 s to 0.60 s decreases mean strict accuracy to 92.361 %. A further reduction to 0.50 s lowers it to 85.222 % and raises the mean maximum overlap depth to 112.90 mm.

The corresponding throughput increases, but this improvement is not evidence of better control:

$$8027.3 \longrightarrow 10360.0 \text{ parcels/h}.$$

It is produced by the higher imposed arrival rate and is accompanied by substantially poorer physical separation. Throughput must therefore be interpreted jointly with accuracy and overlap.

**Downstream treadmill velocity.** Reducing the passive downstream speed produces a more moderate degradation. Mean accuracy remains 96.861 % at 1.60 m/s and 94.361 % at 1.30 m/s.

A slower downstream belt evacuates parcels less rapidly after they leave the controlled AMS region. This increases the time spent in a more congested downstream stream, but within the tested range its effect is less severe than the most aggressive inlet-rate perturbations.

**Batch cardinality and spatial inlet density.** The most severe result is obtained in the `batch3_12` case:

$$\overline{A}_{5\,\mathrm{mm}} = 78.34\,\%, \qquad A_{\mathrm{worst\ agent}} = 72.00\,\%.$$

Its mean maximum overlap depth is 112.8 mm, comparable to the overlap observed for `generation_0p50`.

The comparison between these two cases is particularly informative. Their nominal average arrival rates are equal:

$$\frac{2}{0.50\,\mathrm{s}} = \frac{3}{0.75\,\mathrm{s}} = 4 \text{ parcels/s}. \tag{79}$$

Nevertheless, `batch3_12` performs substantially worse than `generation_0p50`. This supports the hypothesis that the limitation is not determined by temporal inflow rate alone. Simultaneously introducing left, centre, and right parcels produces greater instantaneous spatial occupancy and a central parcel configuration not represented during nominal pairwise training.

This comparison does not constitute a strict causal isolation of spatial density, because the feasible three-parcel environment also uses a reduced maximum side length and a dedicated placement rule. It nevertheless provides strong evidence that inlet structure and simultaneous batch occupancy are important distribution shifts for the learned policy.

**Overall assessment.** The selected PPO configuration is robust in three distinct but limited senses:

- all four independent training runs produce high nominal final accuracy;
- the saved policies generalize well to a doubled parcel count and to rectangular geometry within the tested dimensional limits;
- every tested scenario reaches physical episode completion.

The main operating limitation is dense parcel inflow, particularly when the spatial structure of the inlet differs from the left–right pair distribution used during training. The policy should therefore not be described as universally robust. It is robust within several tested perturbations, but remains sensitive to simultaneous spatial congestion and to high arrival density.

These findings motivate the broader methodological discussion in Section 7, including the role of the nominal training distribution, the accuracy–throughput trade-off, and the limitations of the current kinematic environment.

# 7 Discussion, Limitations, and Future Work

The preceding results establish that a PPO policy can learn effective longitudinal parcel singulation in the validated MATLAB environment. The reference PPO configuration already produces meaningful separation, while the BO-based selection workflow identifies a configuration with substantially higher nominal accuracy, lower dispersion, and smaller residual overlap. The subsequent multiseed and evaluation-only campaigns show that this improvement is not restricted to one isolated saved policy, although performance remains sensitive to the training realization and to specific changes in inlet density.

The purpose of this section is to interpret these findings within the scope of the implemented model. Particular attention is given to the distinction between hyperparameter selection and causal attribution, the relation between reward and physical performance, the operational trade-offs revealed by the selected policy, and the limits of the current robustness evidence.

## 7.1 Interpretation of the policy-selection workflow

The principal outcome of the nominal study is the selection of the `BO_014` PPO hyperparameter configuration. Under the standardized single-seed comparison, it increases the mean final strict edge-gap accuracy from

$$93.44\,\% \quad \text{for the reference PPO configuration to} \quad 99.78\,\%,$$

while reducing both episode-level dispersion and residual overlap.

This result should be interpreted at two distinct levels. First, the standardized comparison identifies the strongest saved nominal agent among the evaluated candidates. Second, the independent multiseed campaign evaluates whether the associated hyperparameter configuration can produce effective policies under different stochastic training realizations.

The multiseed mean strict accuracy of 97.861 %, with every trained agent remaining above 95 %, supports the selection of the complete `BO_014` configuration. However, the individual trained policies are not identical. Their episode-level variability, maximum overlap depth, and collision-correction count differ substantially. The appropriate conclusion is therefore that the selected configuration produces consistently strong final singulation, not that PPO converges to one unique physical control strategy.

The hyperparameter interpretation presented in Section 5.3 must also remain non-causal. Bayesian Optimization modified clip factor, actor and critic learning rates, entropy regularization, discounting, experience horizon, and number of epochs simultaneously. The experiments consequently demonstrate the effectiveness of their joint configuration, but do not identify which individual change is necessary or sufficient for the observed improvement.

A controlled ablation study would be required to establish such causal attribution. For example, changing only the entropy-loss weight while holding all other `BO_014` parameters fixed would test whether reduced explicit exploration is independently responsible for the lower nominal dispersion. That experiment was not part of the present work.

## 7.2 Relationship between training reward and physical performance

One of the main methodological conclusions of the project is that training reward and physical task performance must remain separate.

The instantaneous reward is a shaped learning signal. It is:

- evaluated repeatedly while parcels occupy the downstream region;
- dependent on the number of parcels currently contributing to the reward;
- based on the worst current consecutive gap;
- augmented by a discrete bonus when all current gaps are positive;
- accumulated over an episode whose duration depends on the parcel trajectories.

The final strict accuracy instead evaluates the complete parcel stream once the physical termination condition has been reached. It measures the fraction of consecutive parcel pairs satisfying

$$g > 5\,\mathrm{mm}$$

using parcel boundaries rather than centroid distances.

Consequently, an increase in `AverageReward` does not imply an equivalent increase in final strict accuracy. Similarly, two policies with comparable training-reward plateaus may produce different final overlap depths or collision-correction counts.

The multiseed reward trends illustrate this distinction. The four training trajectories eventually approach broadly comparable reward levels, but their independent physical evaluations remain measurably different. The reward curve is therefore useful for identifying learning progression, instability, or training collapse, but policy selection must rely on the independent edge-based metrics.

A further difficulty is the delayed relation between actions and reward. The policy observes parcels only while their centroids occupy the controllable AMS cells, whereas the reward is evaluated after they enter the passive downstream region. The agent does not directly observe the downstream gaps from which the reward is calculated. It must instead associate earlier local AMS commands with their later effect on separation.

This delayed and partially observed causal structure is compatible with reinforcement learning, but it can increase the variance of the policy-gradient estimate and may contribute to the differences observed across training seeds. It also creates a possible direction for future observation design, provided that any additional information would be physically measurable in the intended application.

## 7.3 Physical and operational trade-offs

The selected policy prioritizes final singulation quality rather than maximum equivalent throughput. In the standardized nominal comparison, `BO_014` achieves the highest strict accuracy and the smallest mean maximum overlap, but its equivalent throughput is lower than those of the reference configuration and `BO_018`.

This does not make throughput irrelevant. It shows instead that throughput cannot be optimized or interpreted independently of the parcel inflow and the resulting physical separation.

The stress campaign provides the clearest example. Reducing the generation period introduces parcels more rapidly and therefore increases the computed equivalent throughput. At the same time, strict accuracy decreases and maximum overlap increases. In particular, the high throughput observed for `generation_0p50` is mainly a consequence of the imposed arrival rate, not evidence that the policy is controlling the parcels more effectively.

A meaningful operating assessment must therefore consider at least:

- final strict edge-gap accuracy;
- episode-to-episode and between-agent variability;
- maximum residual overlap;
- numerical collision-correction count;
- completion time or equivalent throughput;
- non-completion rate.

The collision-correction count requires particular care. It records the number of numerical overlap-resolution operations applied by the simulated environment. A lower value is consistent with cleaner simulated trajectories, but it is not a measurement of contact force, impulse, damage risk, or actual industrial collision severity.

Similarly, the equivalent throughput reported in this work is calculated from a finite transient episode. It allows controlled comparisons between policies and stress conditions, but it should not be interpreted as a direct steady-state capacity measurement for a complete industrial sorting line.

## 7.4 Generalization and tested operating envelope

The stress campaign provides evidence of generalization beyond the nominal training distribution, but only within the tested conditions.

The policy remains close to nominal performance when:

- the number of parcels is increased from 10 to 20;
- square parcels are replaced by rectangles whose width and longitudinal length are independently sampled within the tested dimensional range;
- the downstream treadmill velocity is moderately reduced.

These results indicate that the policy is not restricted to exactly nine final gaps, nor to the nominal equality between parcel width and length. They also show that moderate variation in downstream transport does not immediately invalidate the control strategy learned on the AMS.

The principal operating limitation is instead the density and structure of the inlet flow. Reducing the pair-generation period causes a progressive loss of strict accuracy and an increase in overlap. The deterioration is particularly large at a generation period of 0.50 s.

The `batch3_12` scenario is even more demanding. Although its average arrival rate is equal to that of `generation_0p50`,

$$\frac{3}{0.75\,\mathrm{s}} = \frac{2}{0.50\,\mathrm{s}} = 4 \text{ parcels/s}, \tag{80}$$

it produces a lower mean strict accuracy. This suggests that average temporal rate alone is insufficient to characterize task difficulty. Simultaneous left–centre–right generation creates greater instantaneous spatial occupancy and introduces a central inlet pattern absent from nominal pairwise training.

This comparison must nevertheless remain qualified. The `batch3_12` environment also employs a dedicated placement rule and reduces the maximum square side from 0.40 m to 0.30 m to ensure feasible non-overlapping initialization. Consequently, the test does not isolate the causal effect of batch cardinality. It evaluates robustness to a complete feasible three-parcel inlet configuration.

The tested operating envelope can therefore be summarized as follows:

- strong nominal performance across four independent training seeds;
- good generalization to longer episodes and moderate geometric variation;
- moderate sensitivity to reduced downstream velocity;
- clear sensitivity to high temporal inflow density;
- substantial degradation when the simultaneous spatial inlet pattern differs from the nominal left–right pair structure;
- no non-completion within any tested condition.

The absence of non-completion means that all parcels still reach the final evaluation line within the prescribed horizon. It does not imply successful singulation, since an episode can complete while containing several insufficiently separated or overlapping parcel pairs.

The learned policy should therefore be described as robust over several tested perturbations, but not as universally robust to arbitrary parcel populations, arrival processes, or inlet structures.

## 7.5 Limitations of the current study

The following limitations define the scope of the conclusions.

**Kinematic parcel model.** The environment models parcel motion kinematically. Parcel mass, inertia, friction variability, rotational dynamics, compliance, and detailed contact forces are not represented. The trained policy is therefore validated against the implemented parcel-motion rules, not against a complete dynamic model of the physical AMS.

**Numerical collision handling.** Parcel contacts are managed through numerical overlap-correction rules. These rules are necessary to maintain geometric consistency in simulation, but they can alter the trajectories discontinuously and do not reproduce physical impact or frictional contact. Collision-correction metrics must consequently be interpreted as simulation diagnostics.

**Restricted observation.** The observation contains only cell-local parcel information in the AMS region. Parcels in the passive downstream region are not observed by the policy, even though their gaps determine the reward. Moreover, if multiple parcel centroids occupy the same AMS cell, the fixed cell-based representation cannot encode a complete unordered list of all parcels in that cell. The current representation is compact and compatible with the fixed network input, but it may lose information under dense occupancy.

**Ideal state information.** Parcel positions and dimensions are obtained directly from the simulator. No camera noise, missed detections, occlusions, latency, calibration error, or classification uncertainty is included. The machine-vision and experimental deployment framework developed by Roveda et al. therefore lies outside the scope of this MATLAB project [1].

**Idealized actuation.** The commanded cell velocity and rotation are applied without explicit actuator dynamics, communication delay, saturation transients, rate limits, or control frequency constraints beyond the simulation timestep. Hardware deployment would require these effects to be modelled or measured.

**Project-specific separation threshold.** The strict threshold

$$g > 5\,\mathrm{mm}$$

is used consistently throughout the project to provide a positive clearance margin. It is an evaluation choice for the present study, not a universally validated industrial requirement. Alternative downstream devices may require a larger or task-dependent gap.

**Finite hyperparameter search.** The Bayesian Optimization campaign contains a finite number of objective evaluations over a restricted search space. The fixed training and evaluation seeds improve reproducibility and reduce comparison noise, but they may also favour configurations that perform particularly well on those realizations. The later multiseed and stress campaigns reduce this concern without eliminating it completely.

**Finite training-seed sample.** Four independent training seeds provide meaningful evidence against an isolated successful run, but they are insufficient for a high-precision estimate of the full distribution of PPO outcomes. In particular, the observed range of collision corrections suggests that additional seeds could reveal further variation in transient behaviour.

**Finite stress campaign.** The robustness conclusions apply only to the selected parameter levels and random distributions. The campaign does not map the complete boundary between successful and unsuccessful operation, nor does it evaluate simultaneous changes in several standard stress parameters. With the exception of the dedicated three-parcel case, the tests are primarily one-factor-at-a-time.

**Stochastic policy evaluation.** The agents are evaluated through the implemented MATLAB `getAction` workflow. No separate deterministic evaluator based explicitly on the Gaussian-policy mean was introduced. The reported metrics therefore characterize the reproducible seeded evaluation protocol used in the project, including any stochasticity retained in action selection.

### 7.6 Recommended next experiment

The immediate development priority should be robustness to dense and non-pairwise inlet configurations. This should be addressed through one controlled modification rather than by simultaneously redesigning observation, reward, network architecture, and PPO hyperparameters.

The proposed technical hypothesis is:

> Extending the training distribution to include variable generation periods and occasional feasible three-parcel batches can improve dense-inlet robustness without materially degrading nominal singulation.

The experiment should retain:

- the current 100-dimensional observation;
- the current 50-dimensional action mapping;
- the current reward;
- the current actor and critic architectures;
- the selected `BO_014` PPO hyperparameters;
- the current edge-based evaluation metrics.

Only the training distribution should be modified. A possible curriculum would sample predominantly nominal two-parcel batches while introducing:

- generation periods in a bounded interval containing 0.75 s, 0.60 s, and 0.50 s;
- a limited probability of feasible left–centre–right three-parcel batches.

The principal risk is that broader training variability may reduce nominal performance or cause the policy to learn an overly conservative strategy with lower throughput. Another risk is that the agent may exploit the changed parcel size or placement distributions rather than learning genuinely improved congestion management.

The modification should be accepted only if, over multiple independent training seeds:

1. nominal mean strict accuracy remains within a predefined tolerance of the current multiseed result;
2. nominal overlap and non-completion do not worsen materially;
3. `generation_0p50` and `batch3_12` show higher strict accuracy and lower maximum overlap;
4. the improvement is observed across several trained agents rather than in one isolated seed.

This experiment would directly test the main limitation identified by the current campaign while leaving the validated RL formulation unchanged.

## 7.7 Longer-term extensions

After testing the training-distribution hypothesis, further developments could be introduced progressively.

A first extension would investigate whether limited downstream information can be included in the observation using variables that would also be measurable in a real system. This could reduce the temporal gap between the controlled action and the observed separation outcome.

A second extension would reconsider the reward. Potential improvements include explicit control of reward scale across different parcel counts, prevention of repeated scoring of the same physical separation event, and clearer alignment between the reward bonus and the strict 5 mm evaluation threshold. Any reward modification should be evaluated against the current reward while keeping the environment and PPO configuration fixed.

A third extension would replace the purely kinematic collision model with a higher-fidelity parcel model including rotation, friction, inertia, and contact dynamics. The added model complexity should be justified through a specific discrepancy between simulated and physical trajectories rather than introduced by default.

Finally, transfer toward a physical AMS would require perception uncertainty, state-estimation latency, actuator dynamics, communication delays, control-rate constraints, and safety validation. These elements are central to the experimental framework of Roveda et al., but they were intentionally excluded from the scope of the present simulation project [1].

# 8 Conclusions

This project investigated the use of Proximal Policy Optimization for the continuous control of a $5 \times 5$ Active Motion Surface devoted to longitudinal parcel singulation. The complete workflow was developed in MATLAB using Reinforcement Learning Toolbox and was preceded by a systematic verification of the implemented environment. Observation ordering, action normalization and denormalization, parcel generation, kinematics, lateral boundaries, collision correction, episode termination, and visual consistency were checked before interpreting the PPO training results.

A central methodological outcome of the work is the separation between the reward used for policy learning and the physical quantities used for policy evaluation. The shaped downstream reward provided a sufficiently dense signal for training, but was not treated as direct evidence of task success. Final singulation was instead evaluated from the longitudinal boundaries of consecutive parcels, using a strict project criterion of

$$g > 5\,\mathrm{mm}.$$

This edge-based formulation accounts for the variable parcel dimensions and avoids the physically incomplete classification that would result from centroid distance alone.

The reference PPO configuration already learned a meaningful singulation policy, achieving a mean final strict accuracy of 93.44 %. Its episode-level variability, worst result of 66.67 %, mean maximum overlap depth of 50.0 mm, and frequent numerical collision corrections nevertheless showed that substantial improvement remained possible.

A high-fidelity Bayesian Optimization campaign was therefore used to identify promising PPO configurations. The shortlisted candidates and the reference configuration were subsequently retrained and compared under the same 2000-episode training budget and the same 100-episode nominal evaluation protocol. All three optimized candidates improved the primary accuracy, episode-level dispersion, worst-case result, and residual overlap relative to the reference PPO configuration.

The final retained configuration, denoted `BO_014`, achieved

$$\overline{A}_{5\,\mathrm{mm}} = 99.78\,\%,$$

with an episode-level standard deviation of 1.56 percentage points, a worst evaluated episode of 88.89 %, and a mean maximum overlap depth of only 1.33 mm. This configuration was selected by prioritizing the physical quality and consistency of singulation. The selection does not imply dominance in every operational metric: `BO_018`, for example, retained a higher throughput and a lower collision-correction count. The final decision therefore reflects the project objective rather than a universal ranking independent of the chosen performance priorities.

The reliability of the selected PPO configuration was subsequently assessed through four independent training realizations. Across the corresponding saved agents, the mean of the nominal strict accuracies was

$$\overline{A}_{\mathrm{MS}} = 97.861\,\%,$$

with a between-training-seed standard deviation of 2.217 percentage points. Every trained agent retained a mean accuracy above 95 %, indicating that the final result was not produced by one isolated favourable initialization. However, overlap depth and numerical collision-correction count remained more strongly seed-dependent. The multiseed campaign therefore confirms robust final singulation quality, but not convergence to one unique transient parcel-handling strategy.

The evaluation-only stress campaign further characterized the operating range of the learned policies. Performance remained close to nominal when the total parcel count was doubled and when square parcels were replaced by random rectangles within the tested dimensional limits. Reducing the downstream treadmill velocity caused a moderate degradation, whereas denser inlet conditions were substantially more critical.

Generating parcel pairs every 0.50 s reduced the mean strict accuracy to 85.22 %. The most severe test was the feasible left–centre–right three-parcel inlet configuration, which produced a mean accuracy of 78.34 %. Since this case and the 0.50 s pair-generation case have the same average parcel arrival rate but different instantaneous inlet structures, their comparison supports the hypothesis that simultaneous spatial occupancy is an important limitation of the nominally trained policy. This conclusion remains conditional because the three-parcel environment also requires a dedicated placement rule and a reduced maximum parcel size.

All tested policies completed every nominal and non-nominal rollout within the prescribed evaluation horizon. Episode completion, however, was not sufficient to establish successful singulation: the most congested cases still exhibited substantial reductions in edge-based accuracy and increases in residual overlap. Throughput must likewise be interpreted jointly with separation quality, since higher imposed inflow rates increased the computed parcel rate while degrading the physical output stream.

Overall, `BO_014` is retained as the final PPO configuration of the project. Within the verified MATLAB environment, it combines high nominal edge-based accuracy, low residual overlap, consistency across independent training seeds, and successful task completion under all tested perturbations. The principal remaining limitation is robustness to dense and non-pairwise inlet configurations that differ from the left–right pair distribution used during training.

The most direct continuation of the work would therefore be to broaden the training distribution by introducing variable generation periods and occasional feasible three-parcel batches, while initially keeping the observation, reward, network architecture, physical metrics, and selected PPO hyperparameters unchanged. Such an experiment would test whether dense-inlet robustness can be improved without sacrificing the strong nominal performance already obtained.

Finally, all conclusions are limited to the implemented kinematic simulation. The present results do not establish performance on a physical AMS, where parcel rotation, friction, inertia, contact dynamics, perception uncertainty, actuator dynamics, latency, and hardware constraints would also affect the closed-loop behaviour. The project should therefore be regarded as a validated simulation study and a reproducible basis for subsequent higher-fidelity or experimental investigation.

## References

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